

DNA Structure

Jason Ryan, MD, MPH

DNA

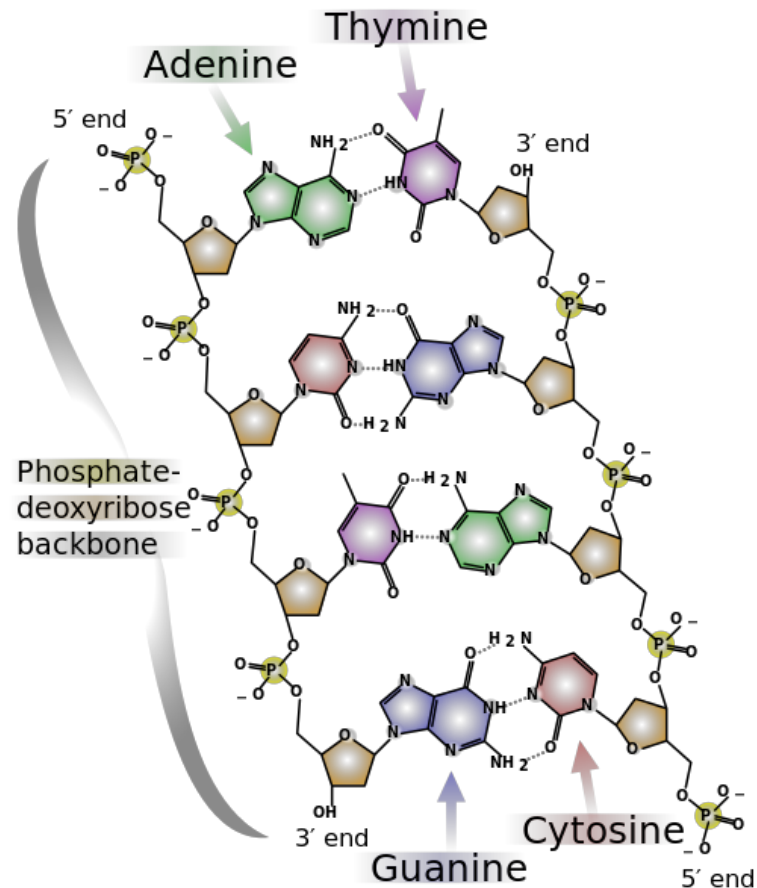
- Contains genetic code
- Nucleus of eukaryotic cells
- Cytoplasm of prokaryotic cells



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DNA Structure

- Sugar (ribose) backbone
- Nitrogenous base
- Phosphate bonds



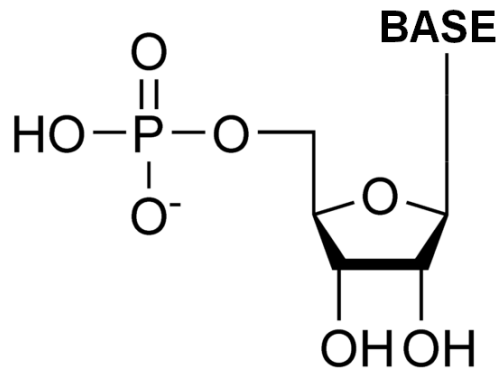
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DNA Vocabulary

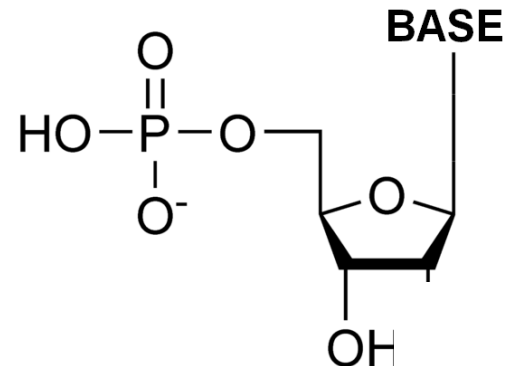
- Nucleotide/Nucleoside
- Nitrogenous base
- Purine/Pyrimidine

Nucleotides

- DNA: Polymer
- Nucleotide: Monomer
 - Pentose sugar
 - Nitrogenous base
 - Phosphate group



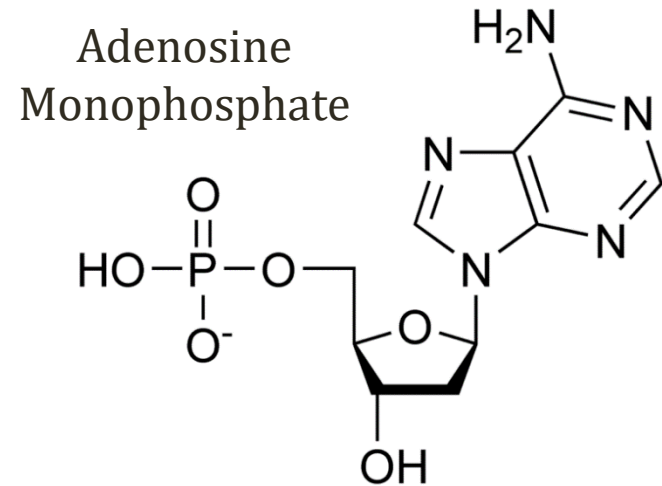
Ribonucleotide



Deoxyribonucleotide

Nucleoside vs. Nucleotide

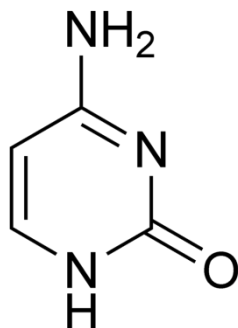
- Nucleotide
 - Nitrogenous base
 - Sugar
 - Phosphate group
- Nucleoside
 - Base and sugar
 - No phosphate group



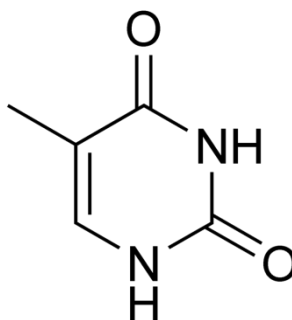
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Nitrogenous Bases

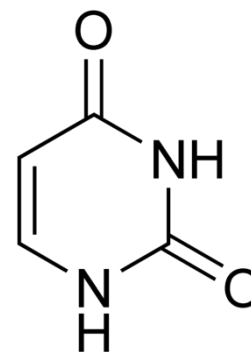
Pyrimidines



Cytosine

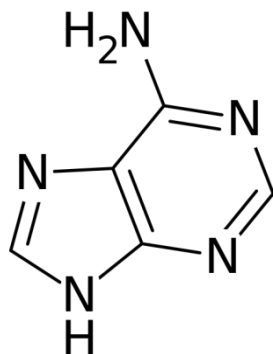


Thymine

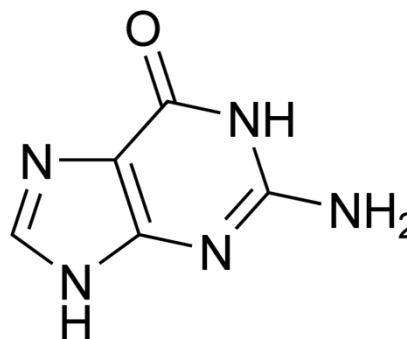


Uracil

Purines

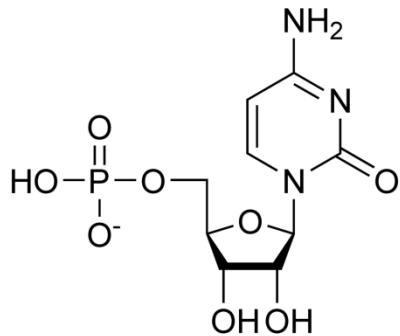


Adenine

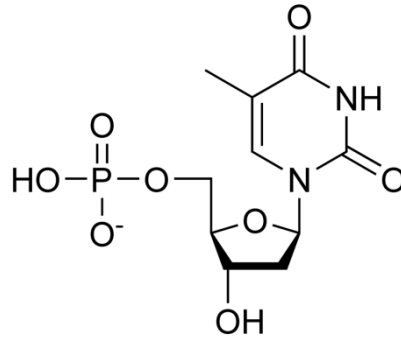


Guanine

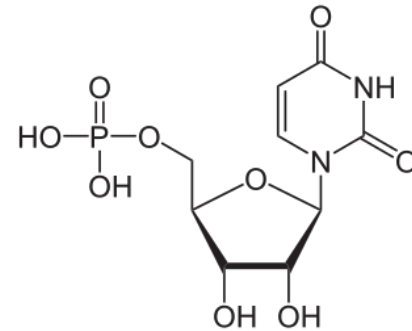
Nucleotides



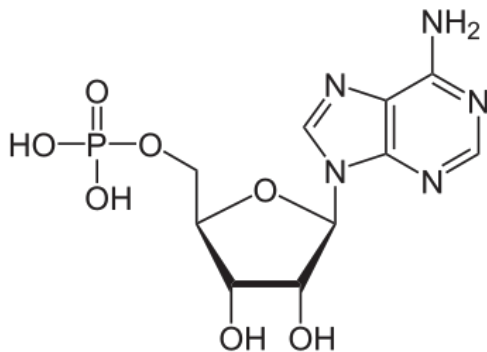
Cytidine



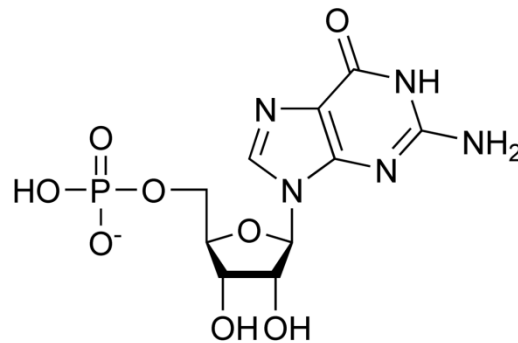
Thymidine



Uridine



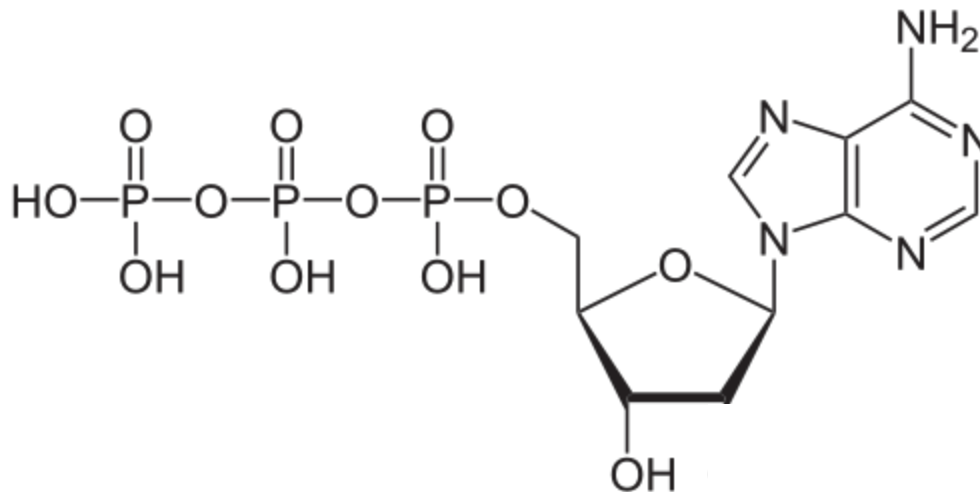
Adenosine



Guanosine

Nucleotides

- Synthesized as monophosphates
- Converted to triphosphate form
- Added to DNA

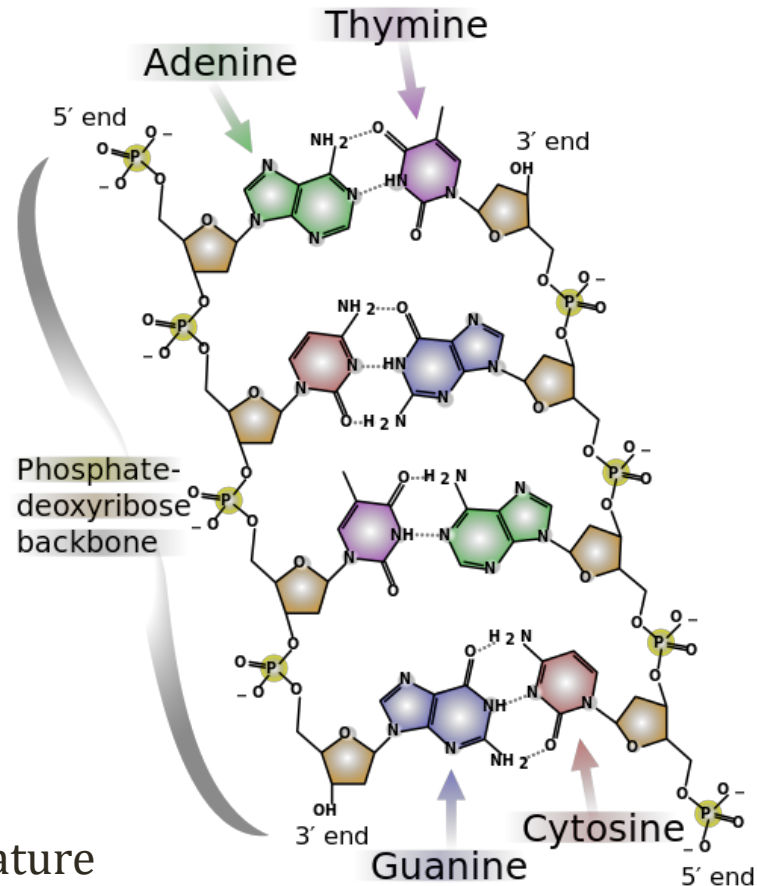


Deoxyadenosine Triphosphate

Base Pairing

- DNA
 - Adenine-Thymine
 - Guanine-Cytosine
- RNA
 - Adenine-Uracil
 - Guanine-Cytosine

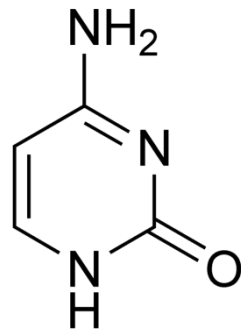
More C-G bonds = $\uparrow$ Melting temperature



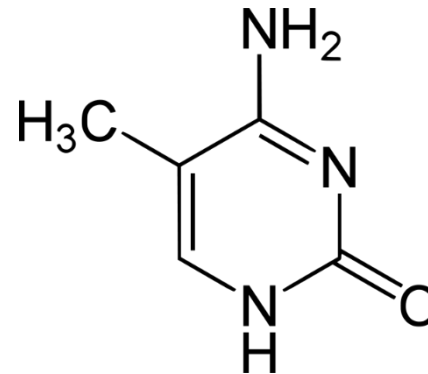
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DNA Methylation

- Methyl group added to cytosine
 - Occurs in segments with CG patterns (“CG islands”)
 - Both strands
- Inactivates transcription (“epigenetics”)
- Human DNA: ~70% methylated
- Unmethylated CG stimulate immune response



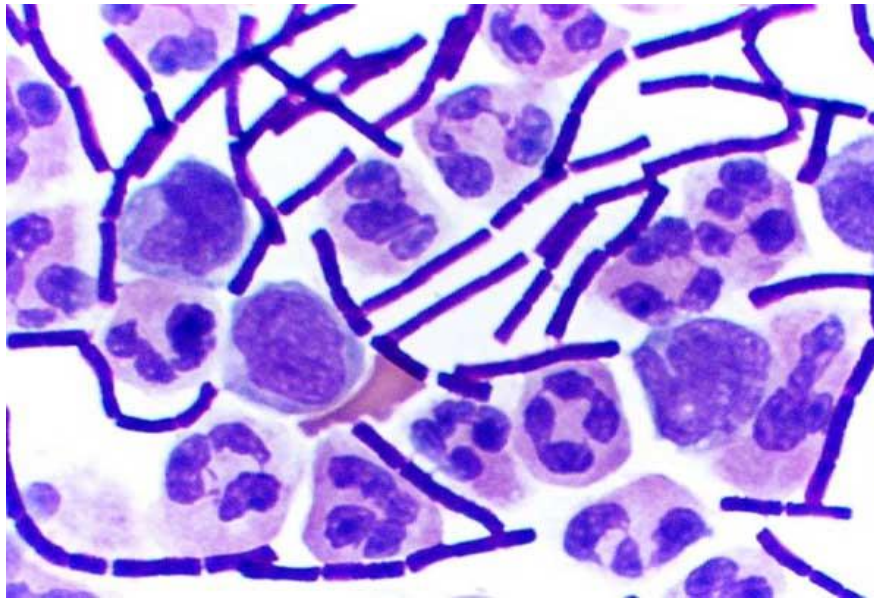
Cytosine



5-methylcytosine

Bacterial DNA Methylation

- Bacteria methylate **cytosine and adenine**
- Methylation protects bacteria from viruses (phages)
- Non-methylated DNA destroyed by endonucleases
- “Restriction-modification systems”

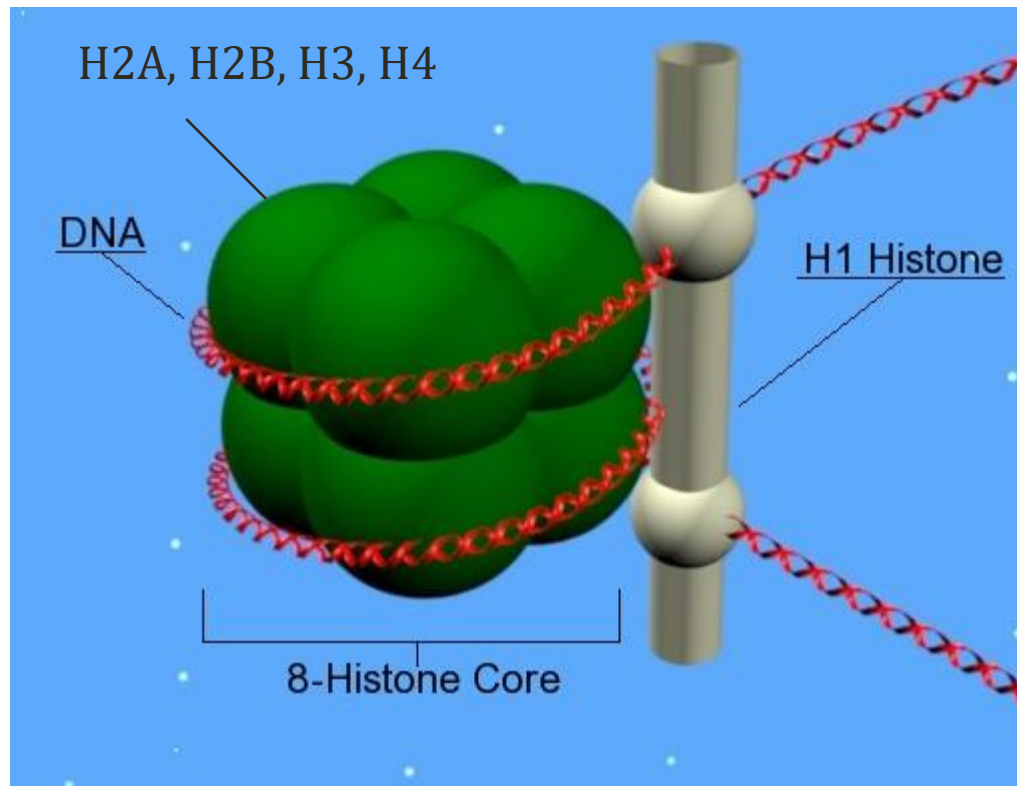


Chromatin

- Found in nucleus of eukaryotic cells
- DNA plus proteins = chromatin
- Chromatin condenses into chromosomes

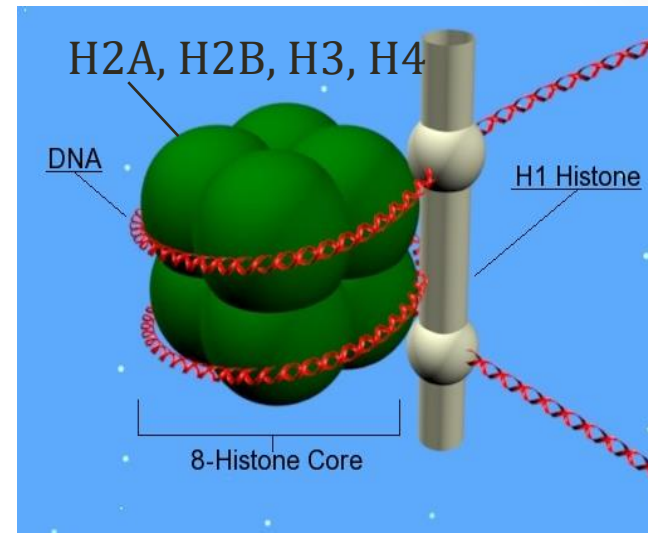
Nucleosome

- Key protein: Histones
- Units of histones plus DNA = nucleosomes



Histones

- Peptides
 - H1, H2A, H2B, H3, H4
- Contain **basic** amino acids
 - High content of lysine, arginine
 - **Positively** charged
 - Binds **negatively** charged phosphate backbone
- H1 distinct from others
 - Not in nucleosome core
 - Larger, more basic
 - Ties beads on string together

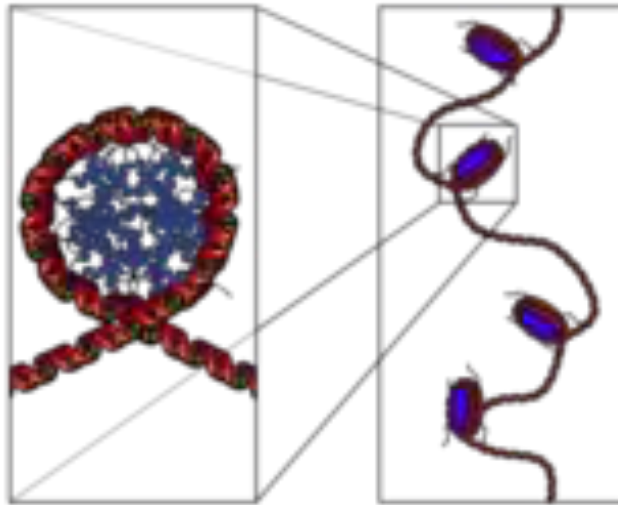


Wikipedia/Public Domain

DNA Structure

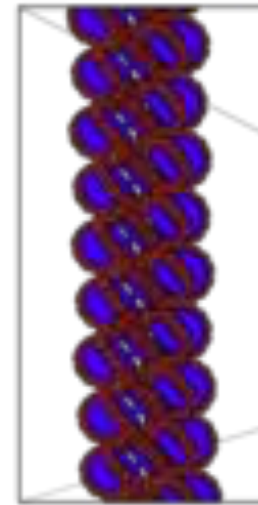


DNA



DNA
plus
Histones

Beads on
a string



H1
Condensation

Richard Wheeler/Wikipedia

Drug-Induced Lupus

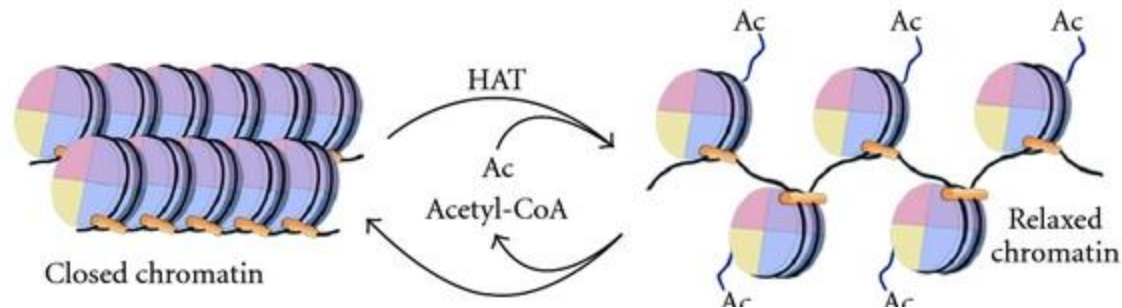
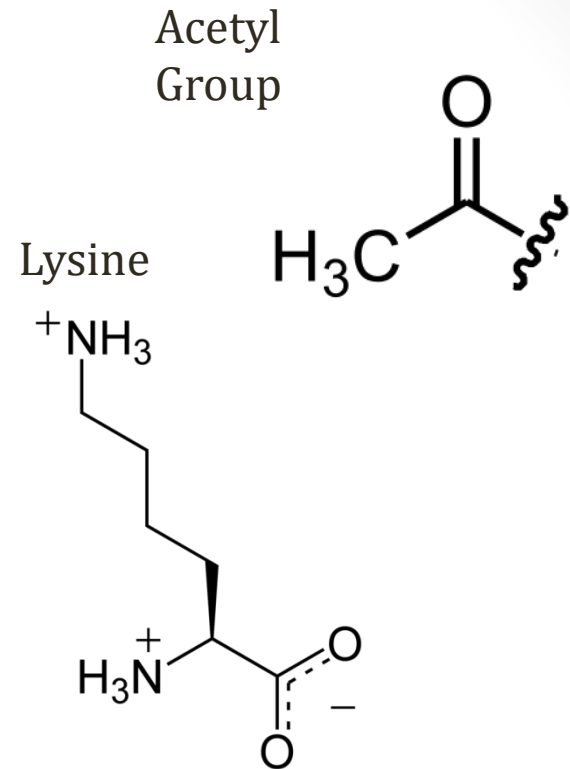
- Fever, joint pains, rash after starting drug
- Anti-histone antibodies (>95% cases)
 - Contrast with anti-dsDNA in classic lupus
- Classic drugs:
 - Hydralazine
 - Procainamide
 - Isoniazid

Chromatin Types

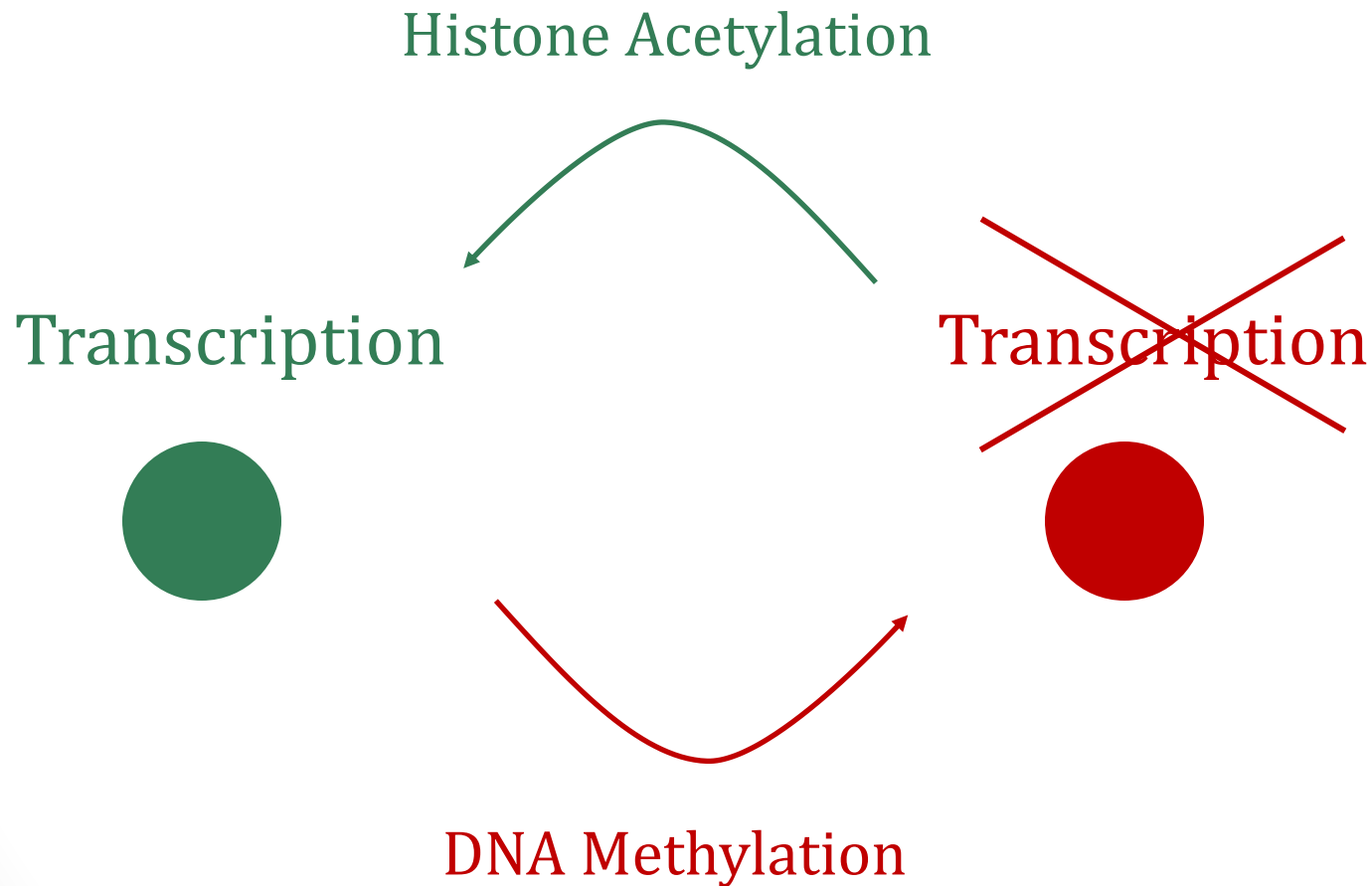
- Heterochromatin
 - Condensed
 - Gene sequences not transcribed (varies by cell)
 - Significant DNA methylation
- Euchromatin
 - Less condensed
 - Transcription
 - Significant histone acetylation

Histone Acetylation

- Acetylation
 - Acetyl group added to **lysine**
 - Relaxes chromatin for transcription
- Deacetylation
 - Reverse effect



Epigenetics



Histone deacetylase inhibitors

HDACs

- Potential therapeutic effects
- Anti-cancer
 - Increased expression of HDACs some tumors
- Huntington's disease
 - Movement disorder
 - Abnormal huntingtin protein
 - Gain of function mutation (mutant protein)
 - Possible mechanism: histone deacetylation → gene silencing
 - Leads to neuronal cell death in striatum

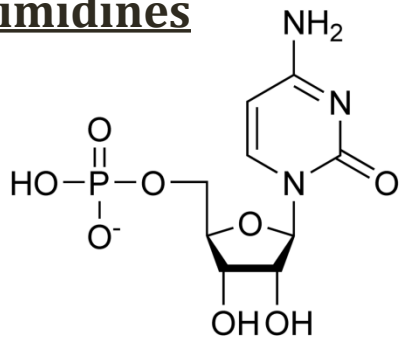
Dokmanovic et al. **Histone deacetylase inhibitors: overview and perspectives**
Mol Cancer Res. 2007 Oct;5(10):981-9.

Purine Metabolism

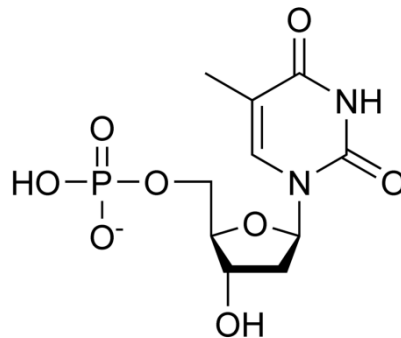
Jason Ryan, MD, MPH

Nucleotides

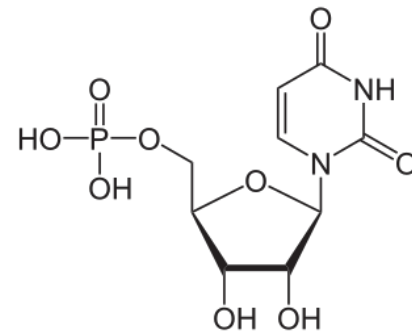
Pyrimidines



Cytidine

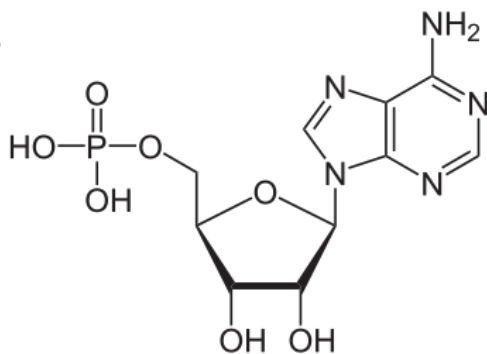


Thymidine

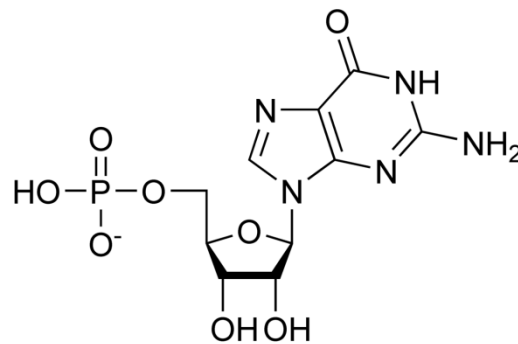


Uridine

Purines



Adenosine



Guanosine

Nucleotide Roles

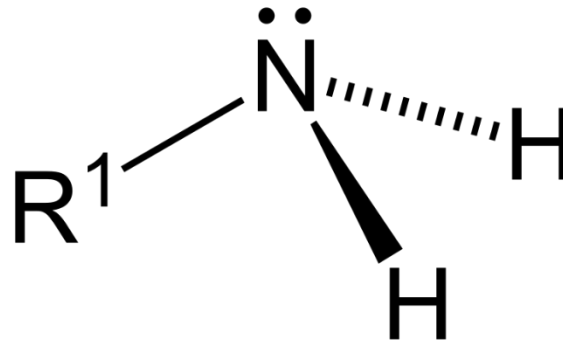
- RNA and DNA monomers
- Energy: ATP
- Physiologic mediators
 - cAMP levels → blood flow
 - cGMP → second messenger

Sources of Nucleotides

- Diet (exogenous)
- Biochemical synthesis (endogenous)
 - Direct synthesis
 - Salvage

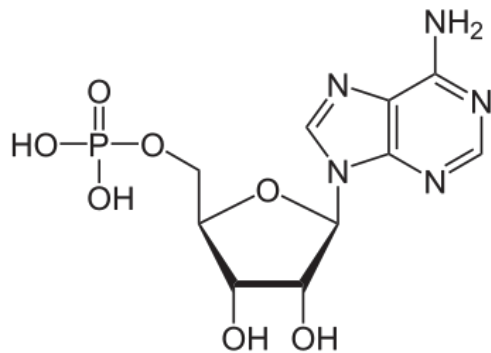
Key Points

- Ribonucleic acids (RNA) synthesized first
- RNA converted to **deoxy**ribonucleic acids (DNA)
- Different pathways for purines versus pyrimidines
- All nitrogen comes from **amino acids**

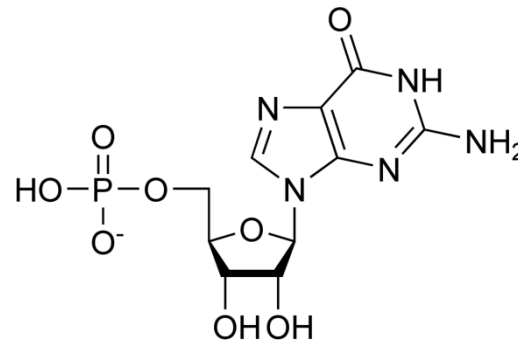


Purine Synthesis

- Goal is to create AMP and GMP
- Ingredients:
 - Ribose phosphate (HMP Shunt)
 - Amino acids
 - Carbons (tetrahydrofolate, CO_2)



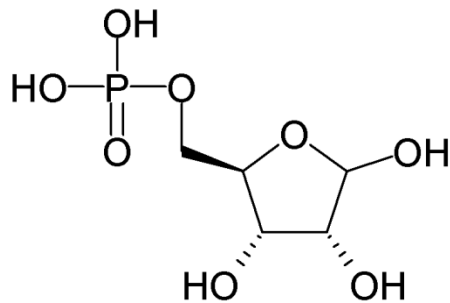
Adenosine



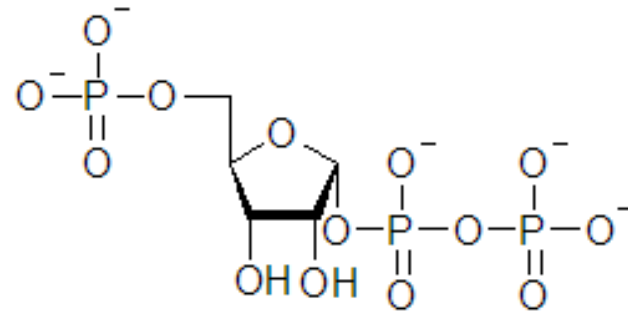
Guanosine

Purine Synthesis

- Step 1: Create **PRPP**



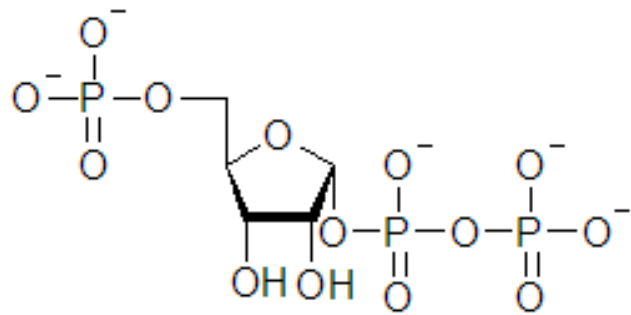
Ribose 5-phosphate



5-Phosphoribosyl-1-pyrophosphate
(PRPP)

Purine Synthesis

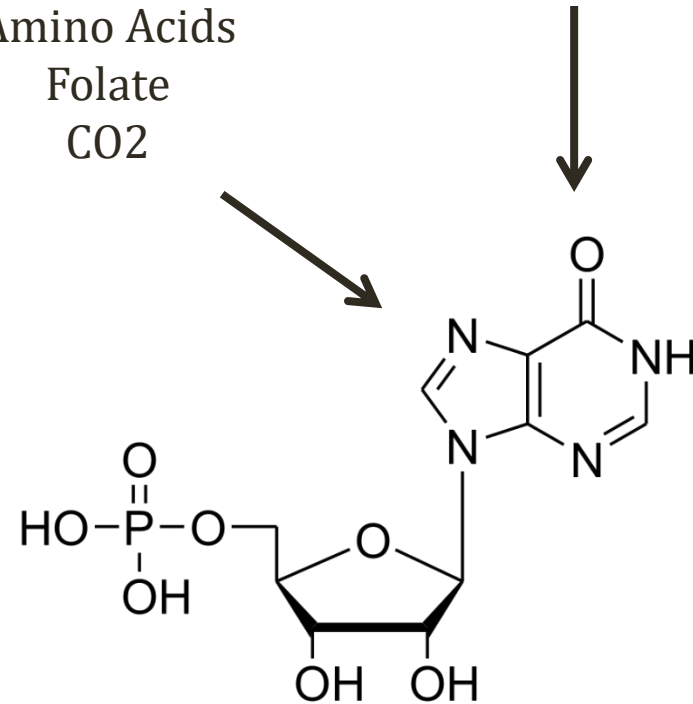
- Step 2: Create **IMP**



5-Phosphoribosyl-1-pyrophosphate
(PRPP)

Amino Acids
Folate
CO₂

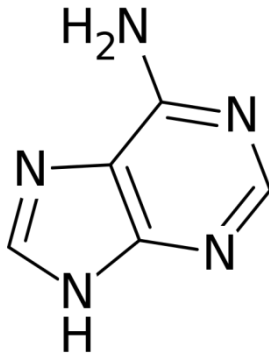
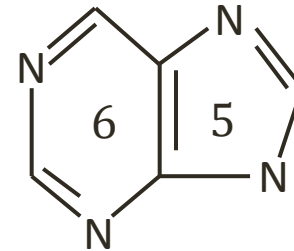
Hypoxanthine



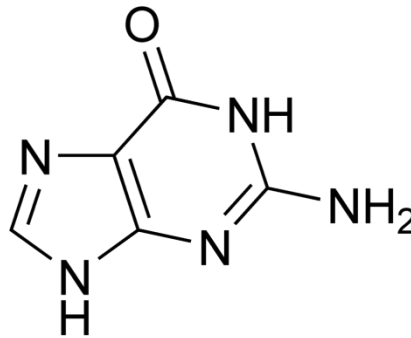
Inosine monophosphate
(IMP)

Purine Synthesis

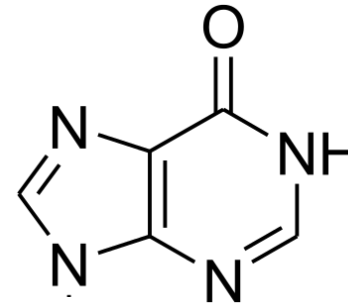
- Two rings with two nitrogens:
 - 6 unit, 3 double bonds
 - 5 unit, 2 double bonds



Adenine



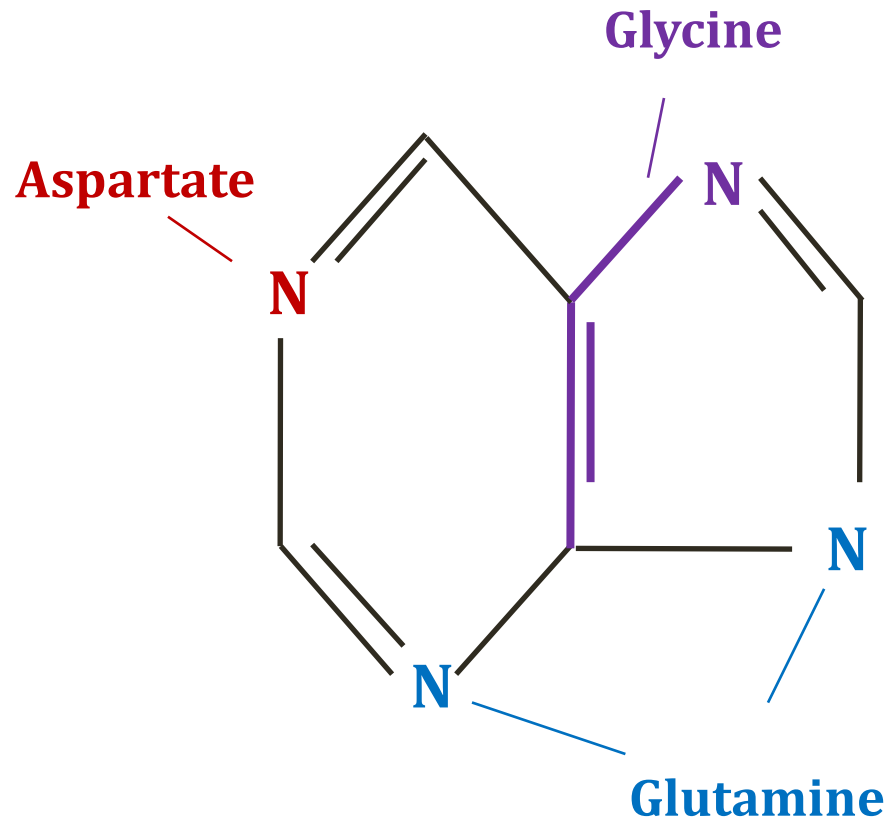
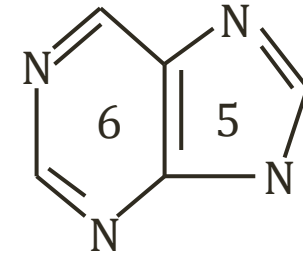
Guanine



Hypoxanthine

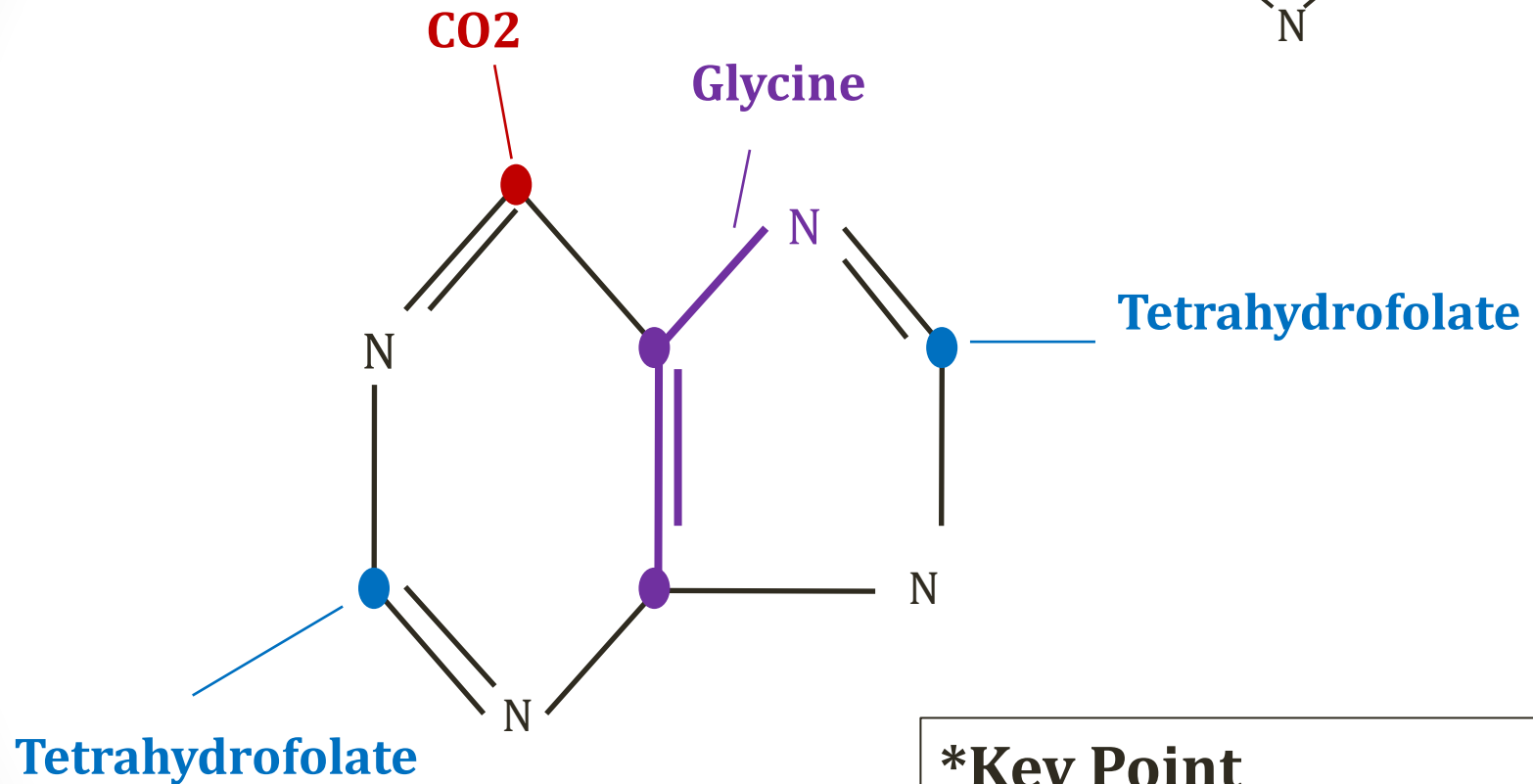
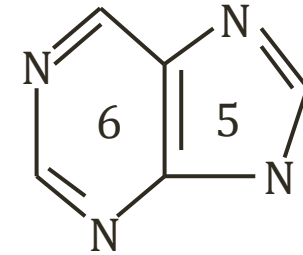
Purine Synthesis

Nitrogen Sources



Purine Synthesis

Carbon Sources

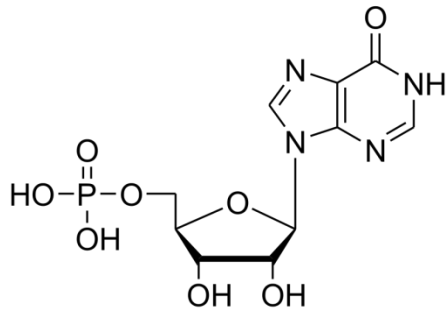


*Key Point

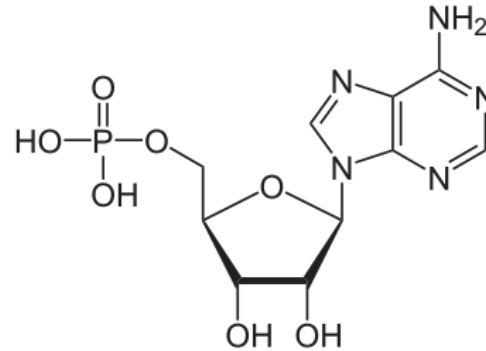
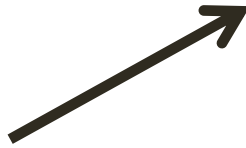
Folate contributes to formation of purines

Purine Synthesis

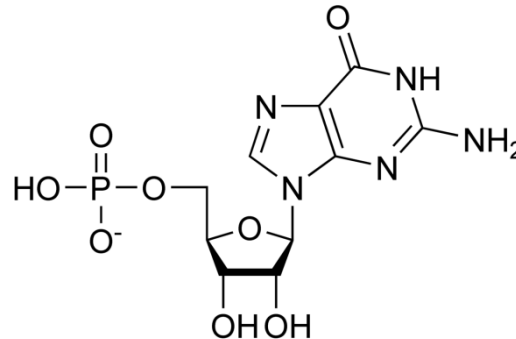
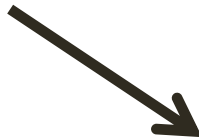
- Step 3: Create AMP and GMP



Inosine monophosphate
(IMP)



Adenosine-MP

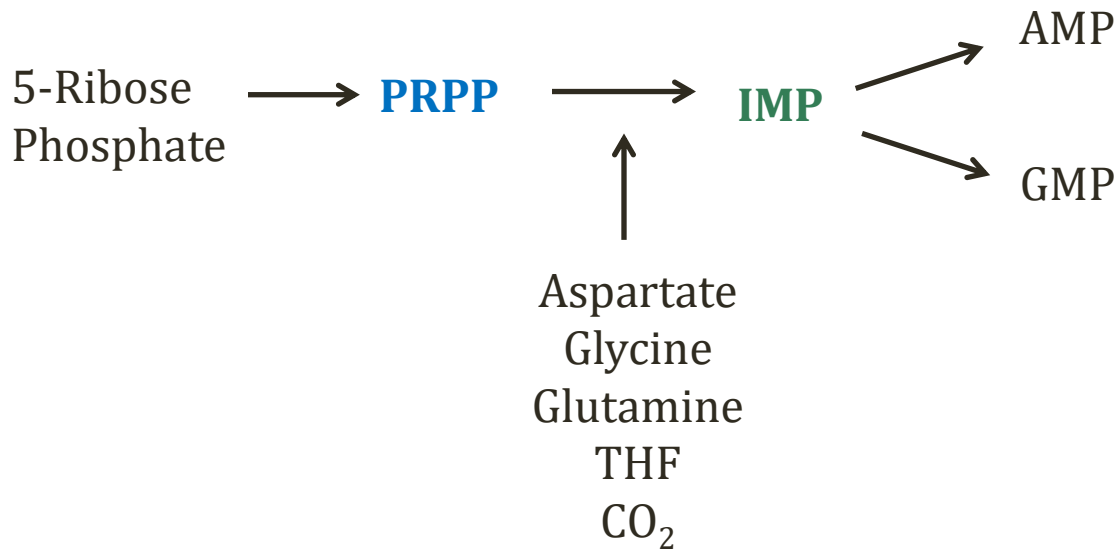


Guanosine-MP

Purine Synthesis

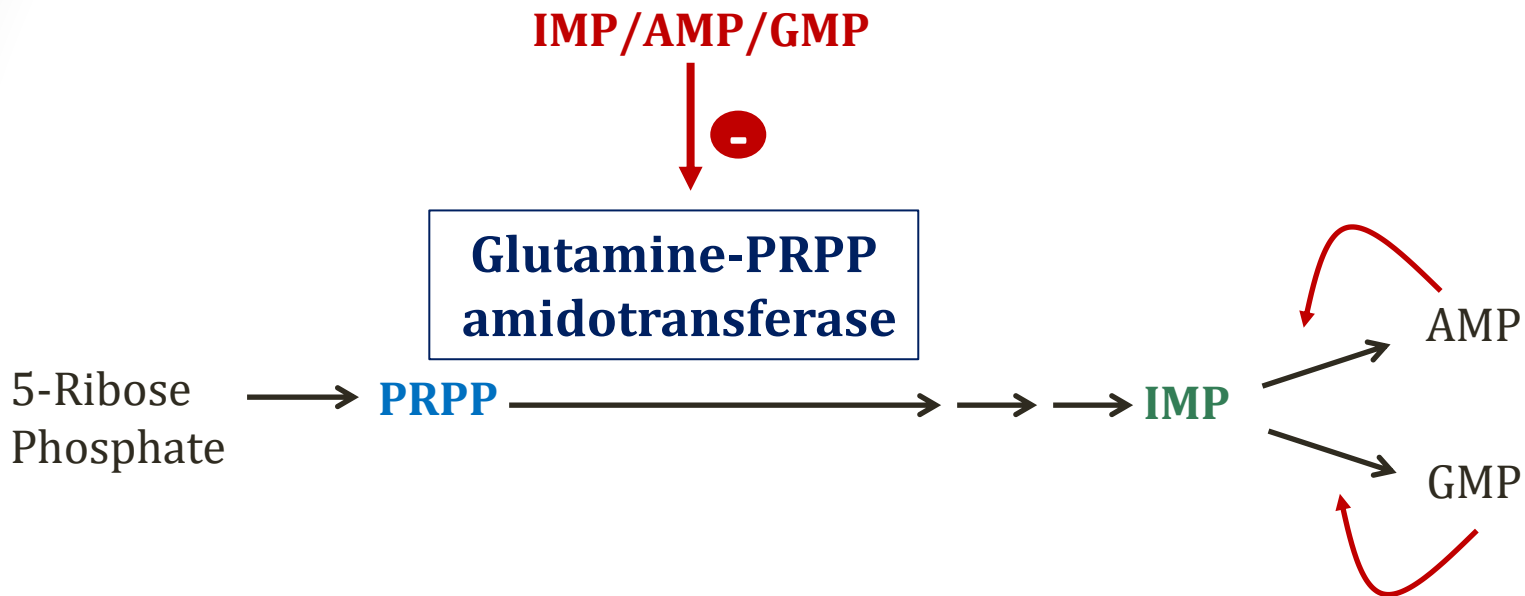
Summary

- Starts with ribose phosphate from HMP shunt
- Key intermediates are **PRPP** and **IMP**

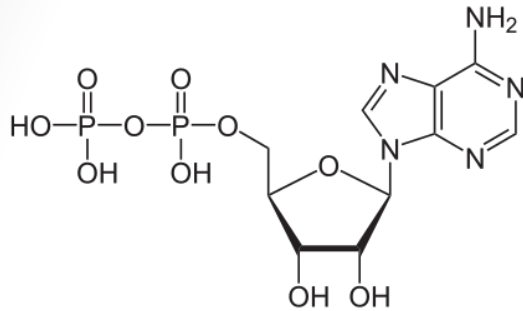


Purine Synthesis

Regulation

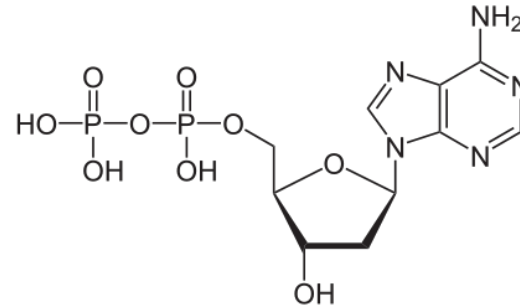


Deoxyribonucleotides

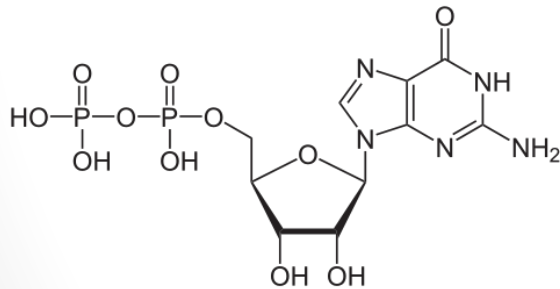


ADP

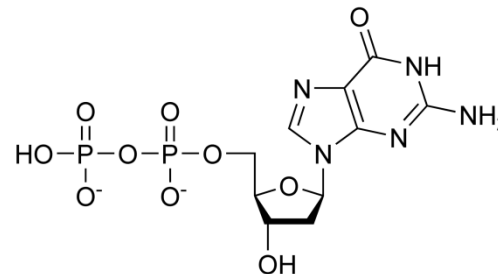
**Ribonucleotide
Reductase**



dADP



GDP



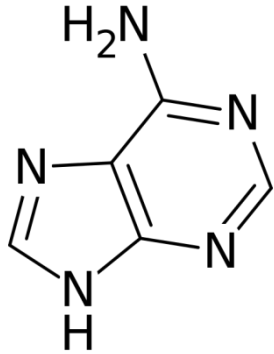
dGDP

Purine Synthesis

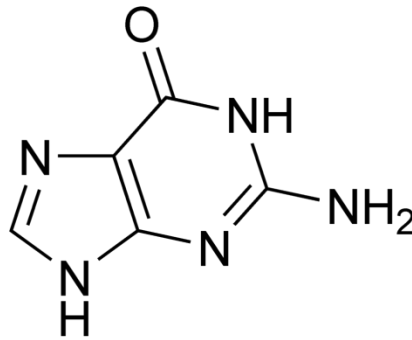
Drugs & Diseases

- Ribavirin (antiviral)
 - Inhibits IMP dehydrogenase
 - Blocks conversion IMP to GMP
 - Inhibits synthesis guanine nucleotides (purines)
- Mycophenolate (immunosuppressant)
 - Inhibits IMP dehydrogenase

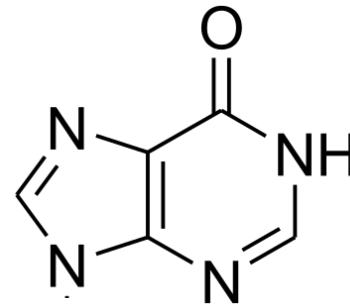
Purine Fates



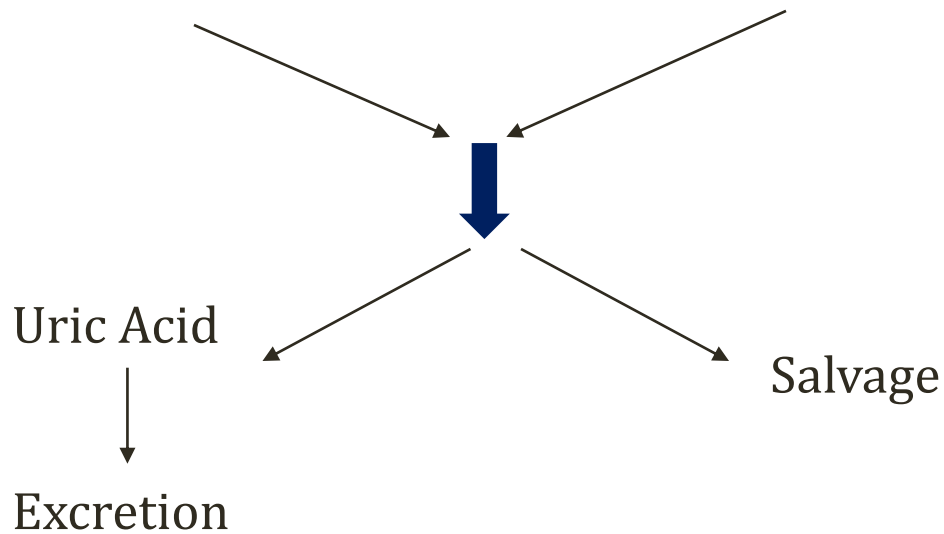
Adenine



Guanine

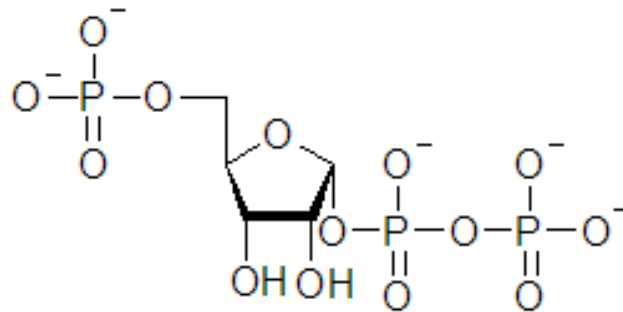


Hypoxanthine



Purine Salvage

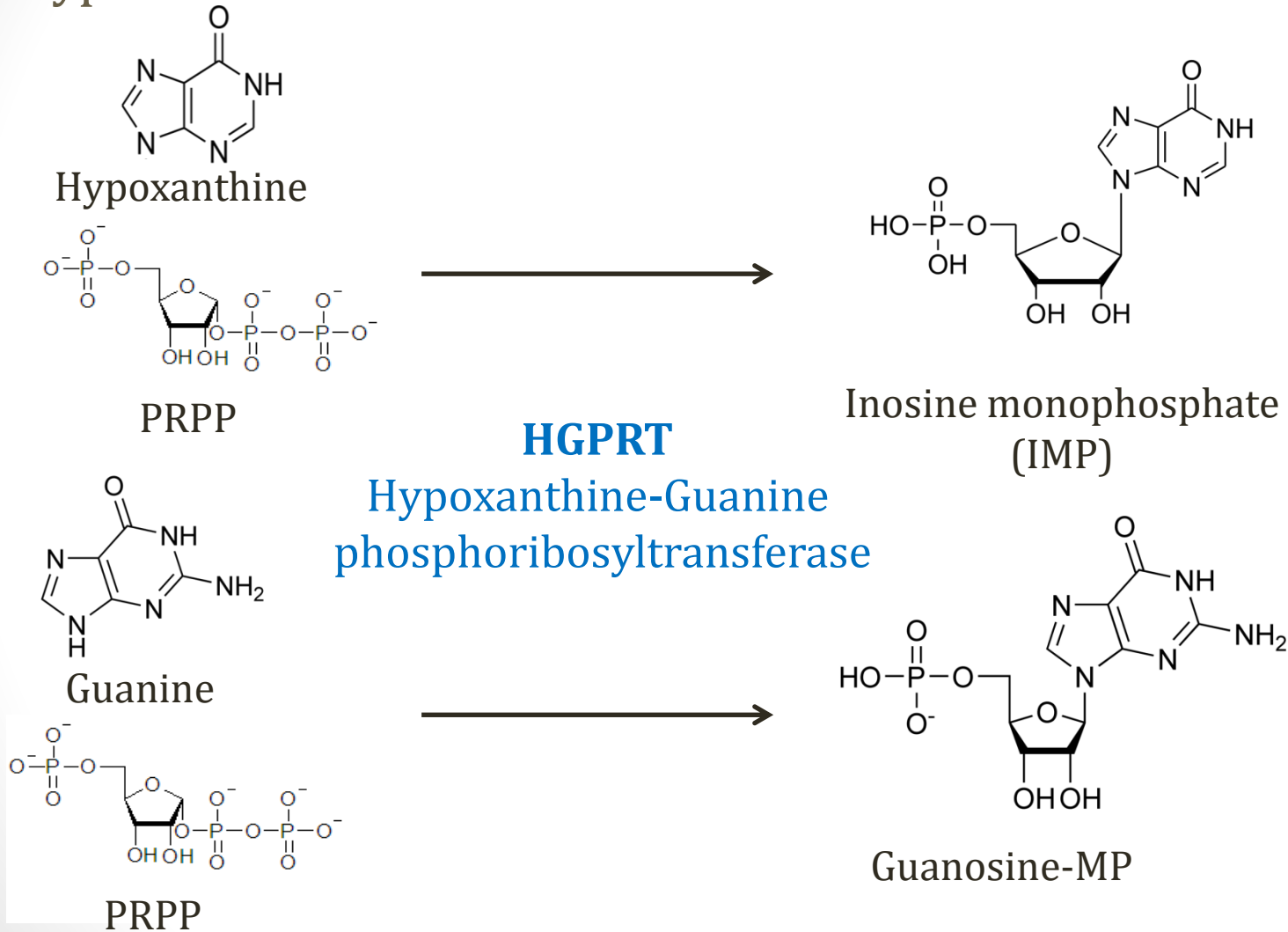
- Salvages bases: adenine, guanine, hypoxanthine
- Converts back into nucleotides: AMP, GMP, IMP
- Requires PRPP



5-Phosphoribosyl-1-pyrophosphate
(PRPP)

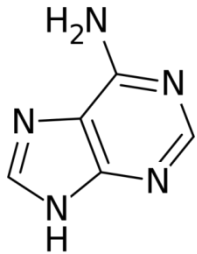
Purine Salvage

Hypoxanthine and Guanine

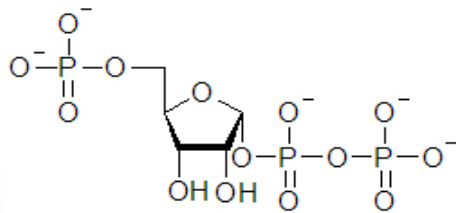


Purine Salvage

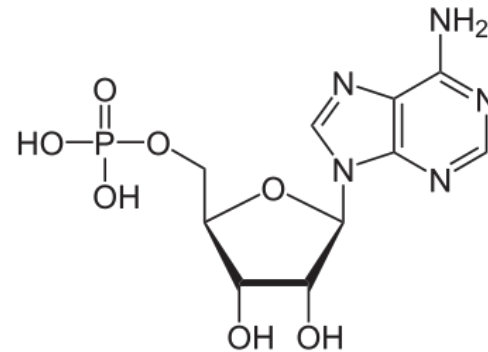
Adenine



Adenine



PRPP



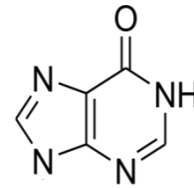
Adenosine-MP

APRT
Adenine
phosphoribosyltransferase

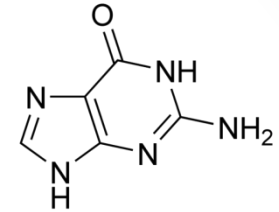
Purine Salvage

Drugs & Diseases

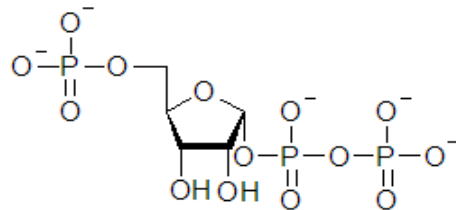
- 6-Mercaptopurine
 - Chemotherapy agent
 - Mimics hypoxanthine/guanine
 - Added to PRPP by HGPRT → Thioinosinic acid
 - Inhibits multiple steps in de novo synthesis
 - ↓IMP/AMP/GMP



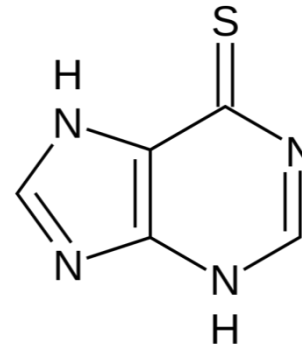
Hypoxanthine



Guanine



PRPP

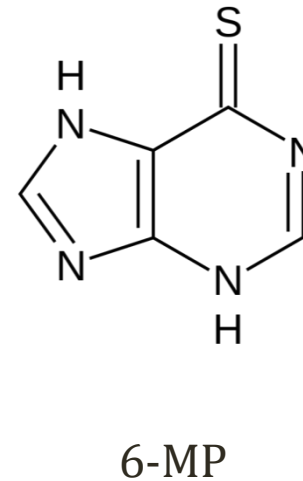
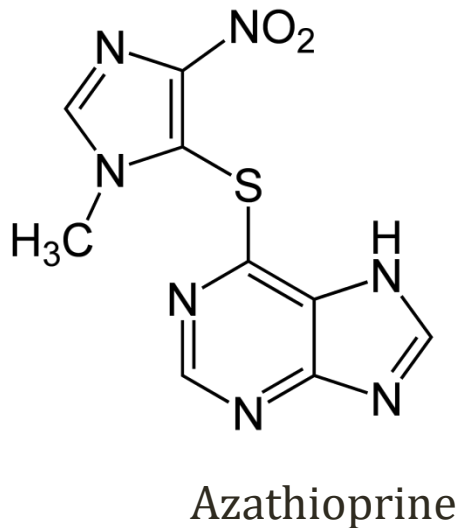


6-MP

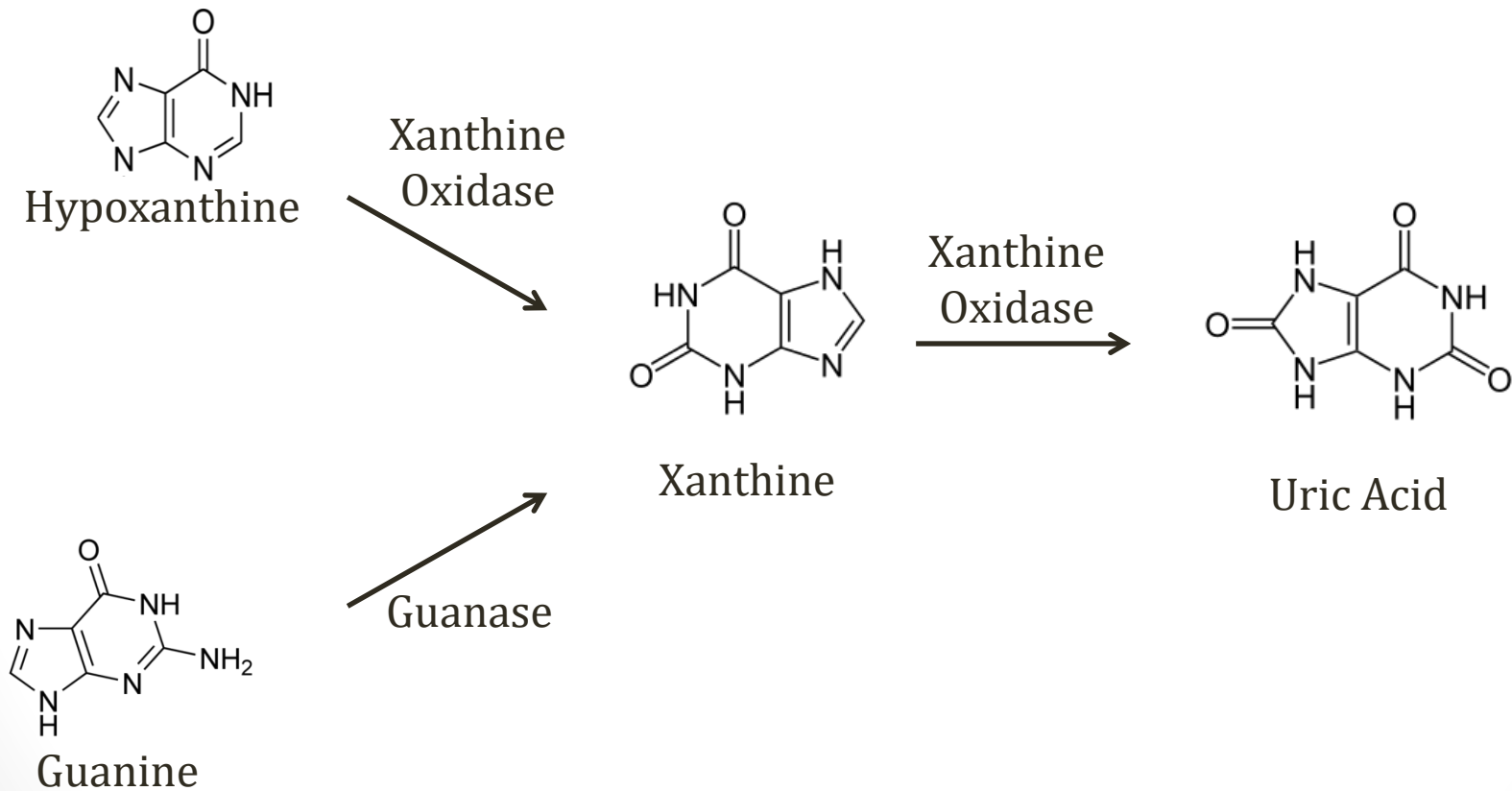
Purine Salvage

Drugs & Diseases

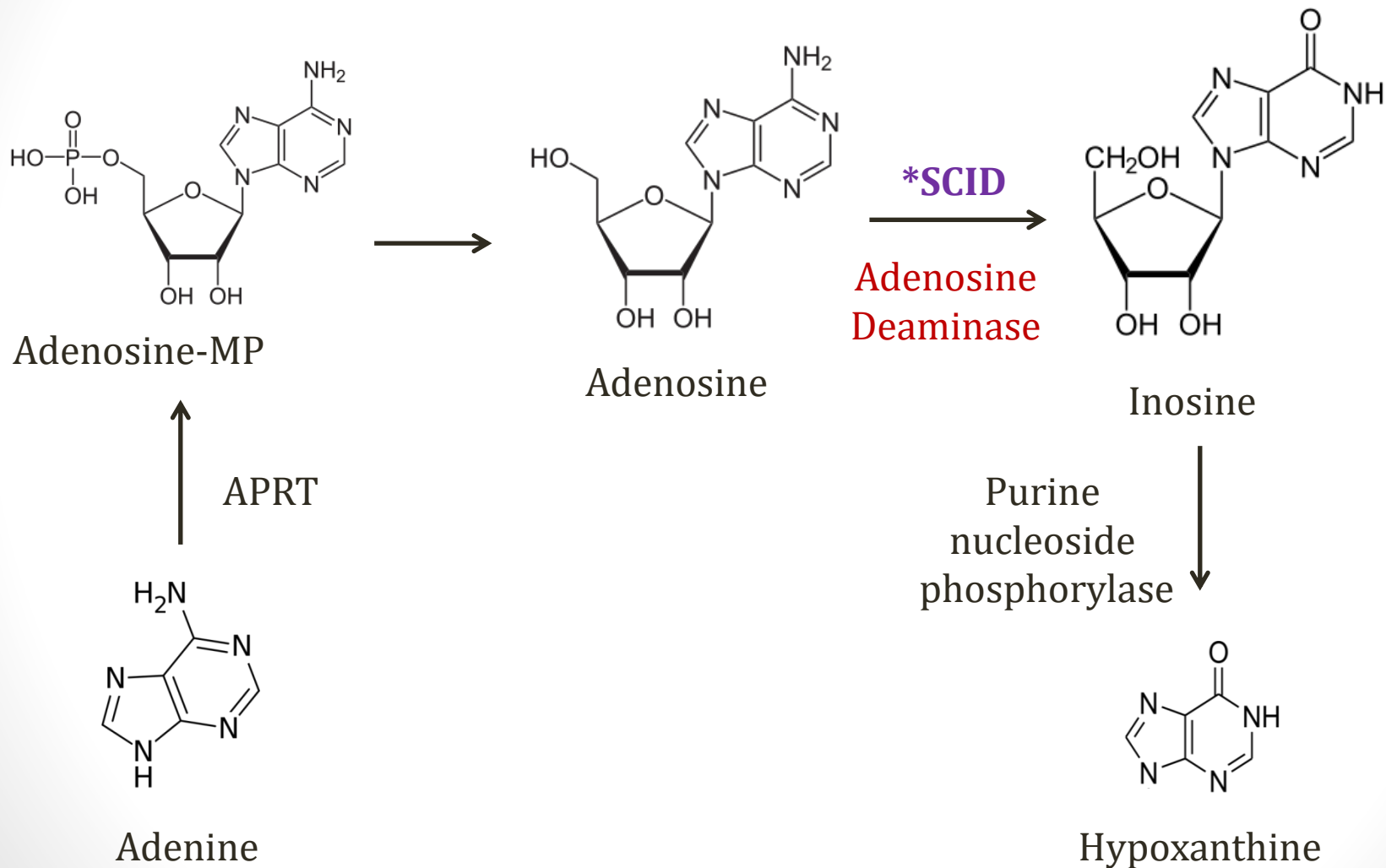
- Azathioprine
 - Immunosuppressant
 - Converted to 6-MP



Purine Breakdown

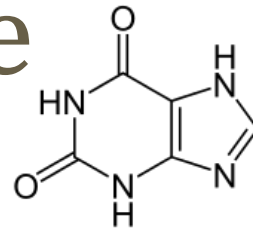


Purine Breakdown



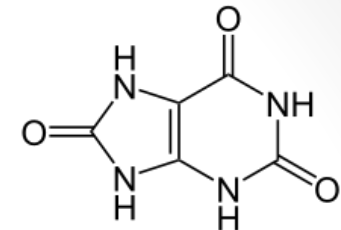
Purine Salvage

Drugs & Diseases



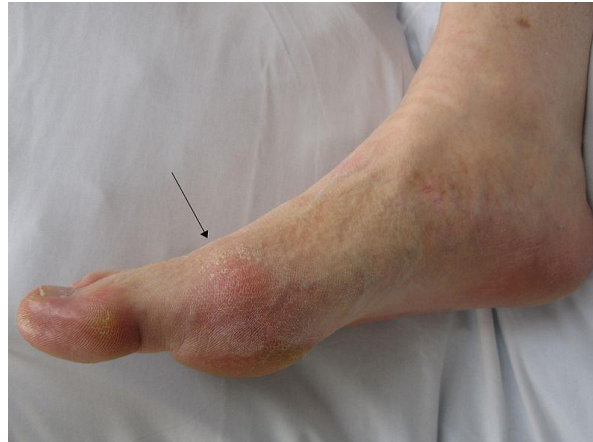
Hypoxanthine

Xanthine
Oxidase
→



Uric Acid

- Gout
 - Excess uric acid
 - Crystal deposition in joints → pain, swelling, redness
 - Can occur from overproduction of uric acid
 - High cell turnover (trauma, chemotherapy)
 - Consumption of purine-rich foods (**meat**, **seafood**)
 - Treatment: inhibit xanthine oxidase (**allopurinol**)

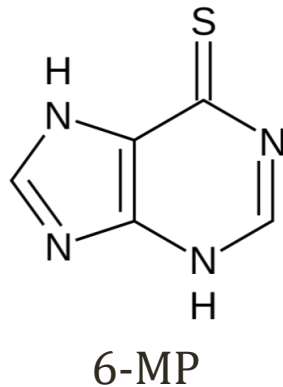


James Heilman, MD/Wikipedia

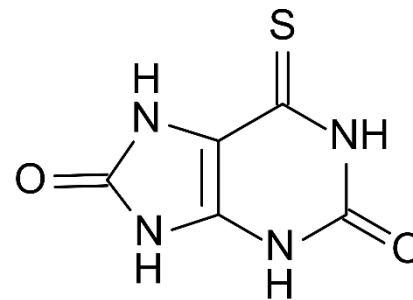
Purine Salvage

Drugs & Diseases

- Azathioprine and 6-MP
 - Metabolized by **xanthine oxidase**
 - Caution with allopurinol
 - May boost effects
 - May increase toxicity



Xanthine
Oxidase
→



Thiouric acid
(inactive)

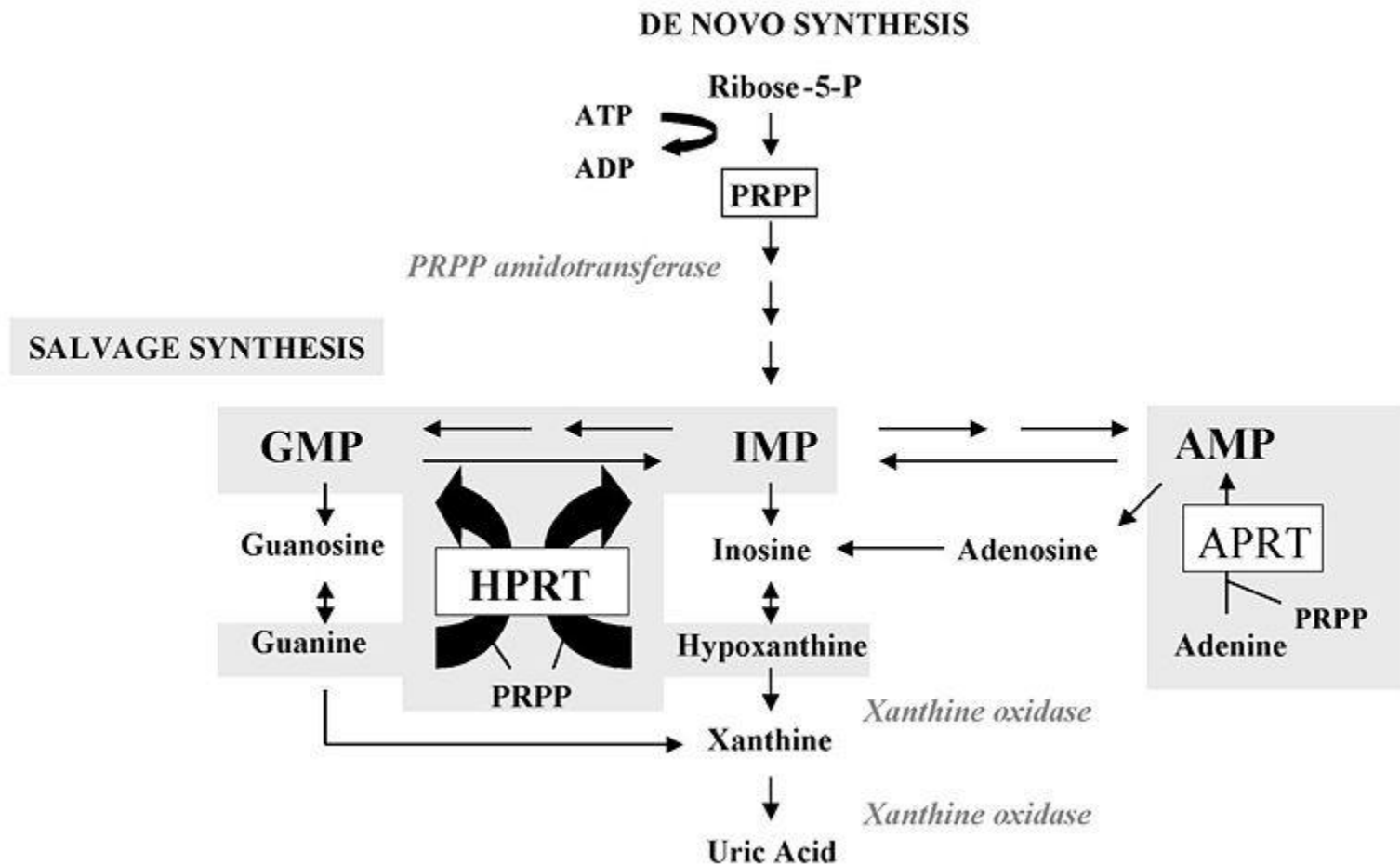
Purine Salvage

Drugs & Diseases

- Lesch-Nyhan syndrome
 - **X-linked absence of HGPRT**
 - **Excess uric acid** production (“juvenile gout”)
 - Excess de novo purine synthesis (**↑PRPP, ↑IMP**)
 - Neurologic impairment (mechanism unclear)
 - Hypotonia, chorea
 - Classic feature: self mutilating behavior (biting, scratching)
 - No treatment
- Classic presentation
 - Male child with motor symptoms, self-mutilation, gout

Purine Metabolism

Summary

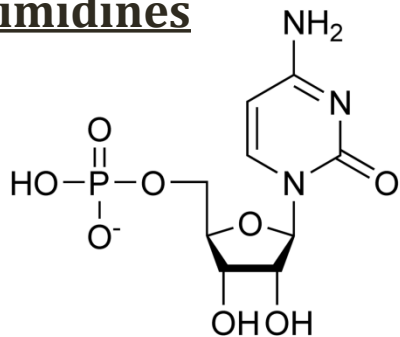


Pyrimidine Metabolism

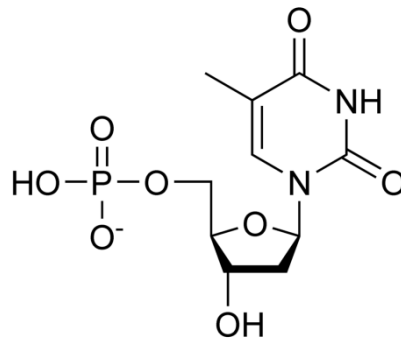
Jason Ryan, MD, MPH

Nucleotides

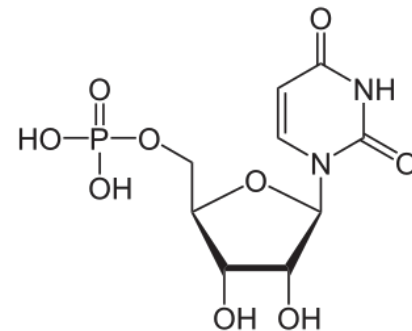
Pyrimidines



Cytidine

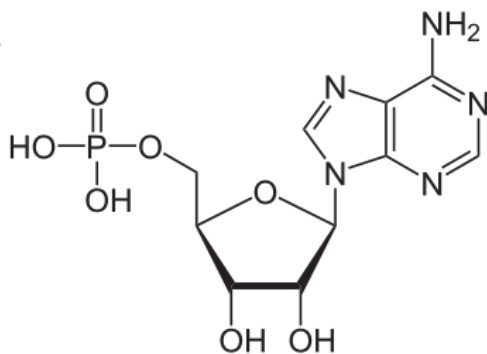


Thymidine

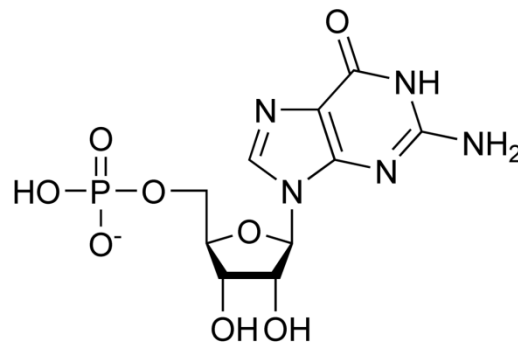


Uridine

Purines



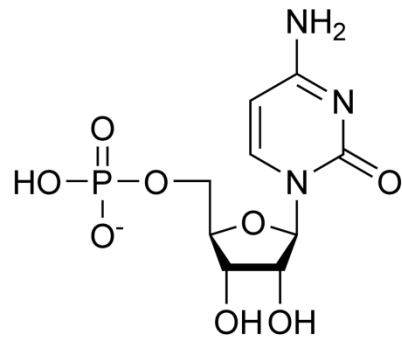
Adenosine



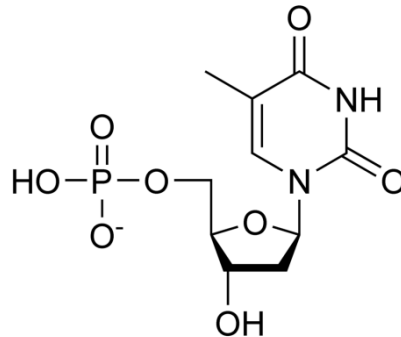
Guanosine

Pyrimidine Synthesis

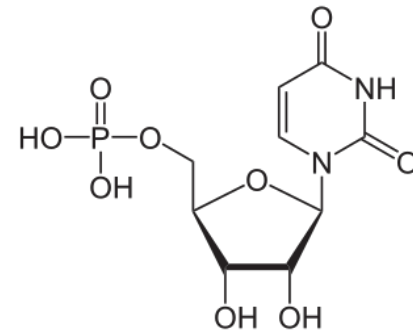
- Goal is to create CMP, UMP, TMP
- Ingredients:
 - Ribose phosphate (HMP Shunt)
 - Amino acids
 - Carbons (tetrahydrofolate, CO_2)



Cytidine



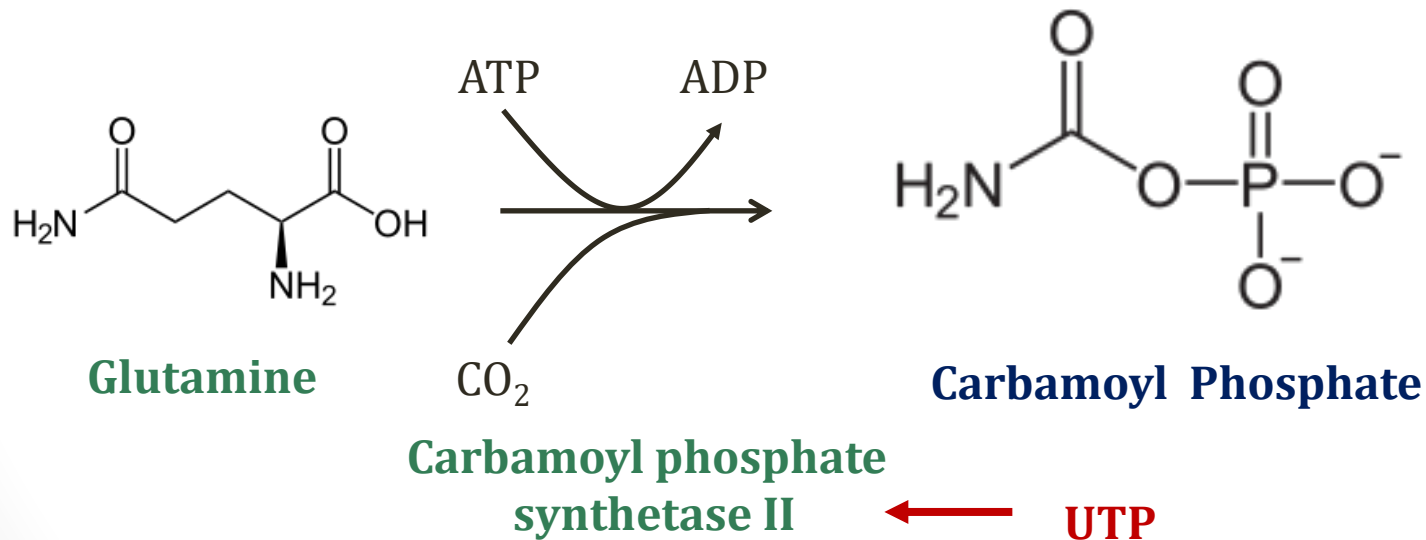
Thymidine



Uridine

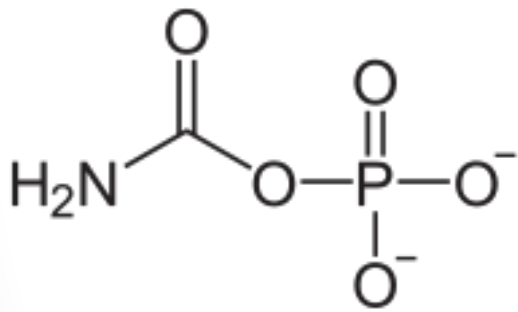
Pyrimidine Synthesis

- Step 1: Make carbamoyl phosphate
- Note: ring formed first then ribose sugar added



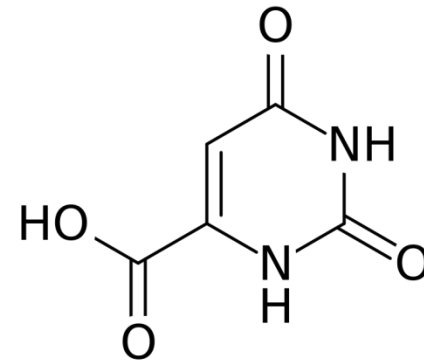
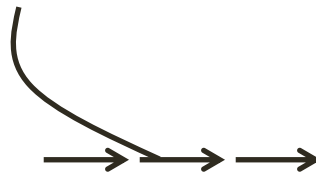
Pyrimidine Synthesis

- Step 2: Make orotic acid



Carbamoyl Phosphate

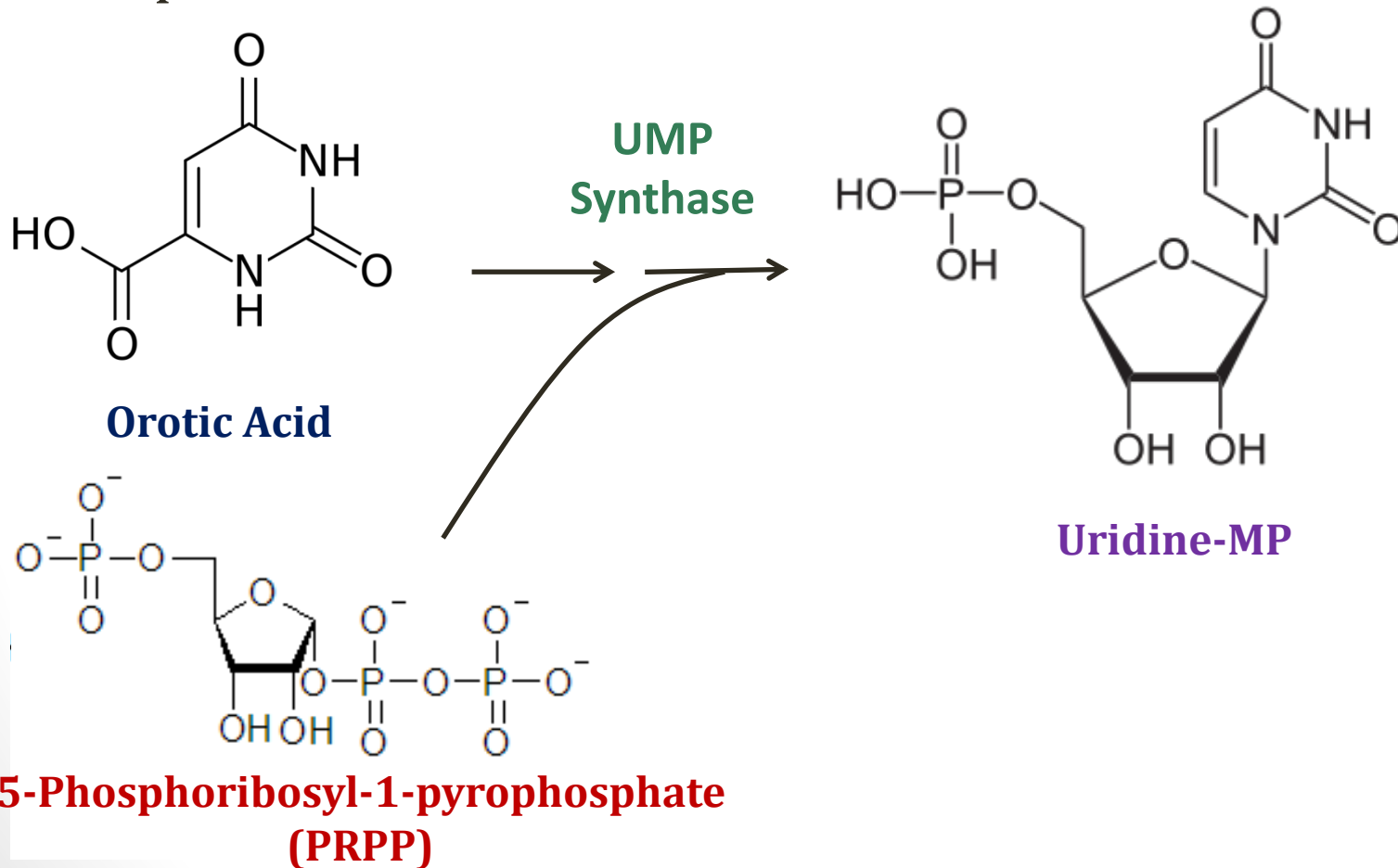
Aspartate



Orotic Acid

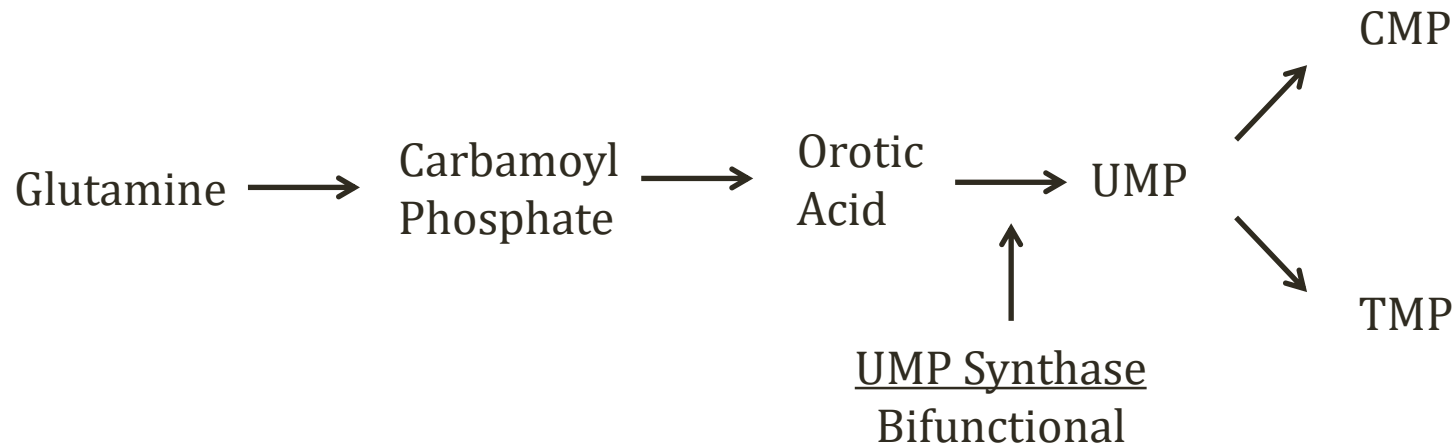
Pyrimidine Synthesis

- Step 3: Make UMP



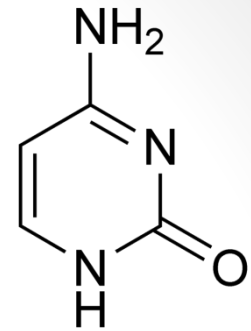
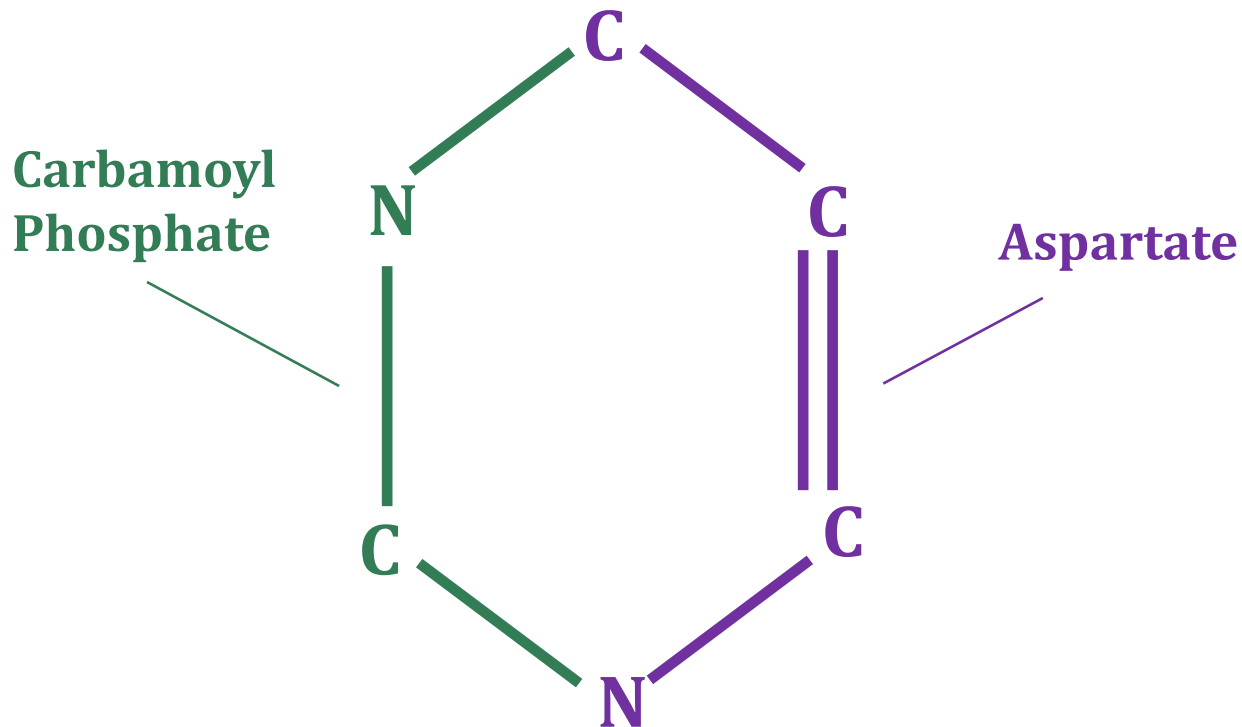
Key Point

- UMP synthesized first
- CMP, TMP derived from UMP

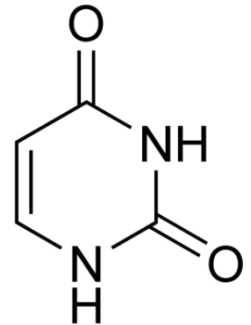


Pyrimidine Ring

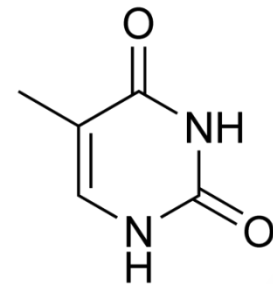
Two nitrogens/four carbons



Cytosine



Uracil

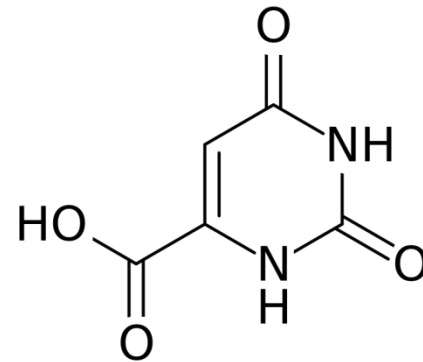


Thymine

Pyrimidine Synthesis

Drugs and Diseases

- **Orotic aciduria**
 - Autosomal recessive
 - Defect in **UMP synthase**
 - Buildup of orotic acid
 - Loss of pyrimidines

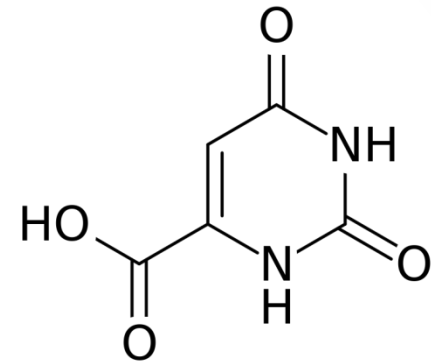


Orotic Acid

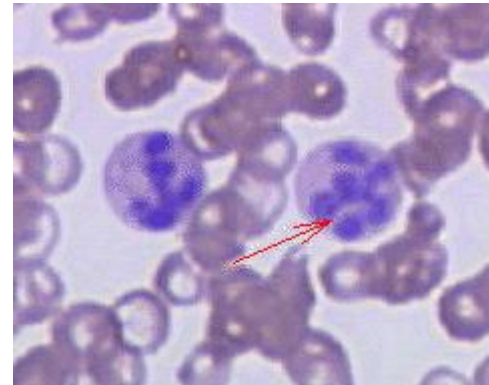
Pyrimidine Synthesis

Drugs and Diseases

- Key findings
 - Orotic acid in urine
 - Megaloblastic anemia
 - No B12/folate response
 - Growth retardation
- Treatment:
 - **Uridine**
 - Bypasses UMP synthase



Orotic Acid



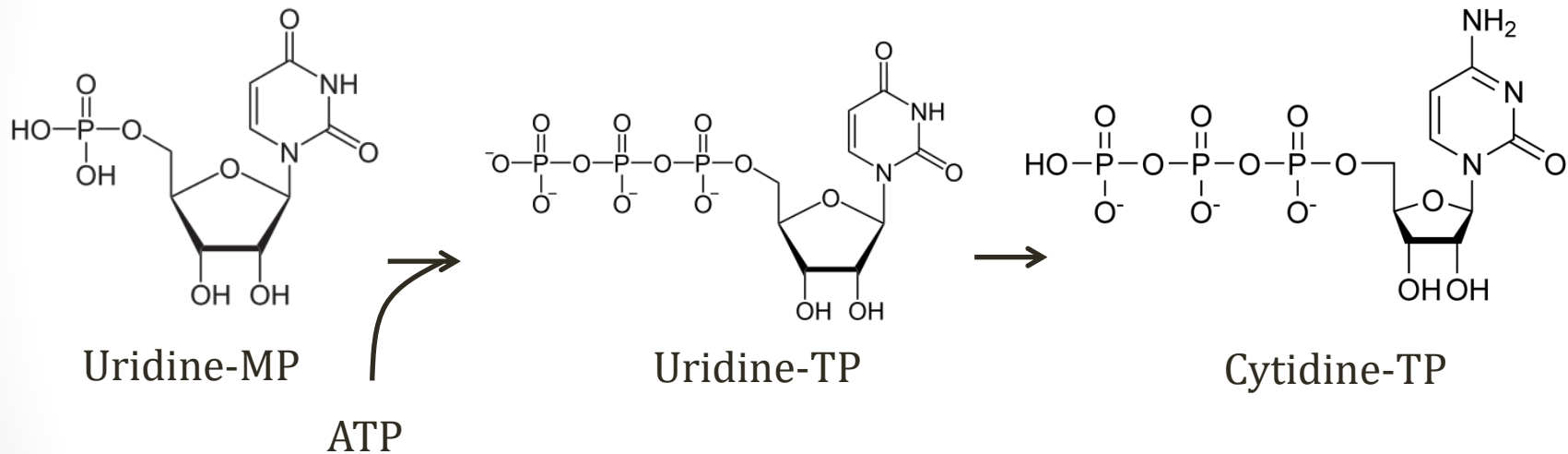
Megaloblastic Anemia

Ornithine transcarbamoylase

OTC

- Key urea cycle enzyme
- Combines **carbamoyl phosphate** with ornithine
- Makes citrulline
- OTC deficiency → increased **carbamoyl phosphate**
- ↑ carbamoyl phosphate → ↑ orotic acid
- Don't confuse with orotic aciduria
 - Both have orotic aciduria
 - OTC only: ↑ **ammonia levels** (urea cycle dysfunction)
 - Ammonia → encephalopathy (baby with lethargy, coma)

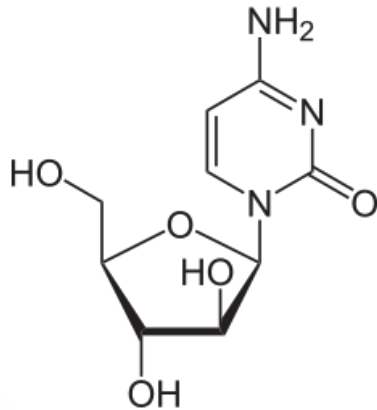
Cytidine



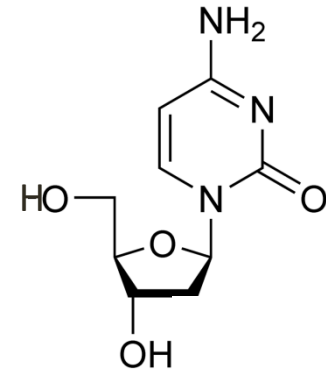
Pyrimidine Synthesis

Drugs and Diseases

- Ara-C (Cytarabine or cytosine arabinoside)
 - Chemotherapy agent
 - Converted to araCTP
 - Mimics dCTP (pyrimidine analog)
 - Inhibits DNA polymerase



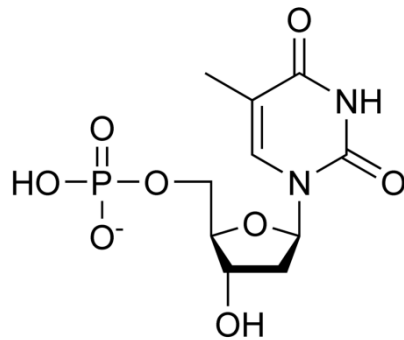
Ara-C



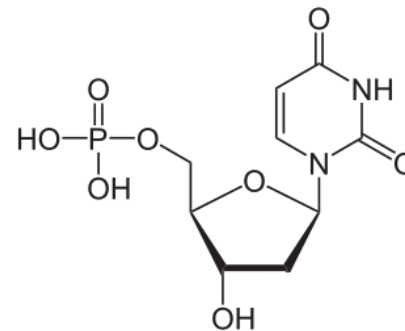
dCytidine

Thymidine

- Only used in DNA
- **Deoxy**thymidine is only required nucleotide
- Synthesized from deoxyuridine



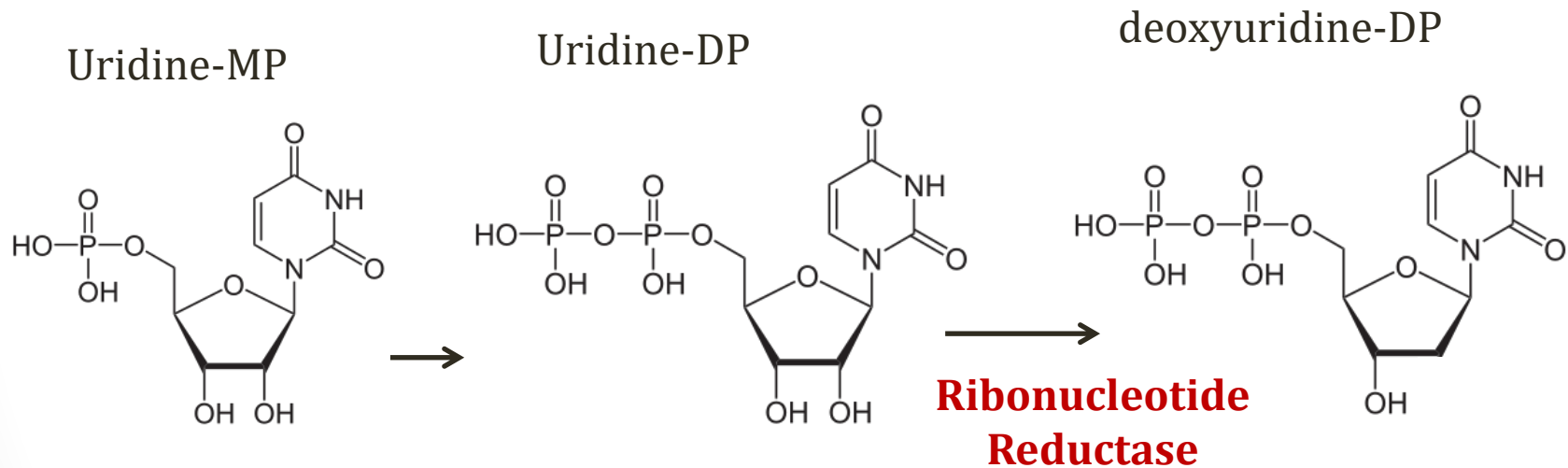
Thymidine



Uridine

Thymidine

- Step 1: Convert UMP to dUDP



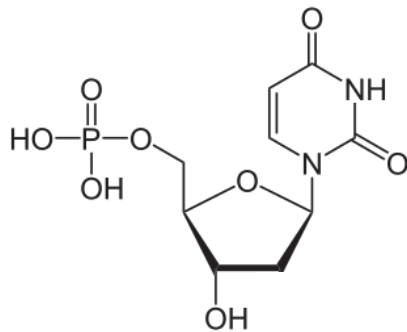
Pyrimidine Synthesis

Drugs and Diseases

- Hydroxyurea
 - Inhibits **ribonucleotide reductase**
 - Blocks formation of deoxynucleotides (RNA intact!)
 - Rarely used for malignancy
 - Can be used for polycythemia vera, essential thrombocytosis
 - Used in sickle cell anemia
 - Causes increased fetal hemoglobin levels (mechanism unclear)

Thymidine

- Step 2: Convert dUDP to dUMP
- Step 3: Convert dUMP to dTMP

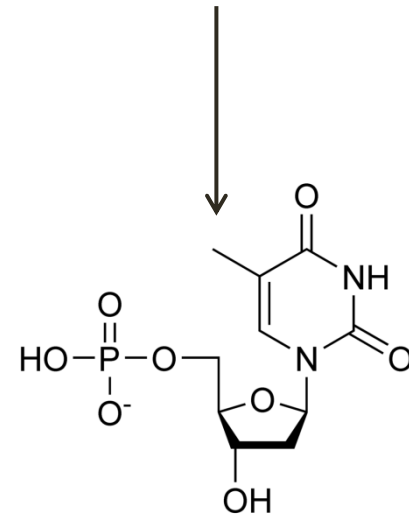


deoxyuridine-MP
(dUMP)

**Thymidylate
Synthase**

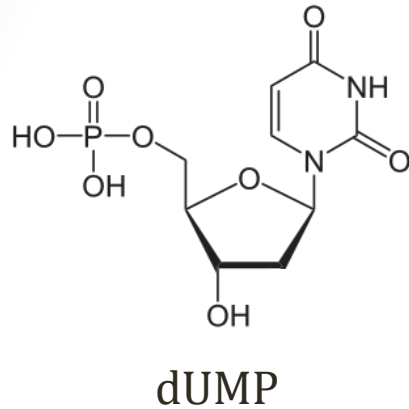


1 Carbon added

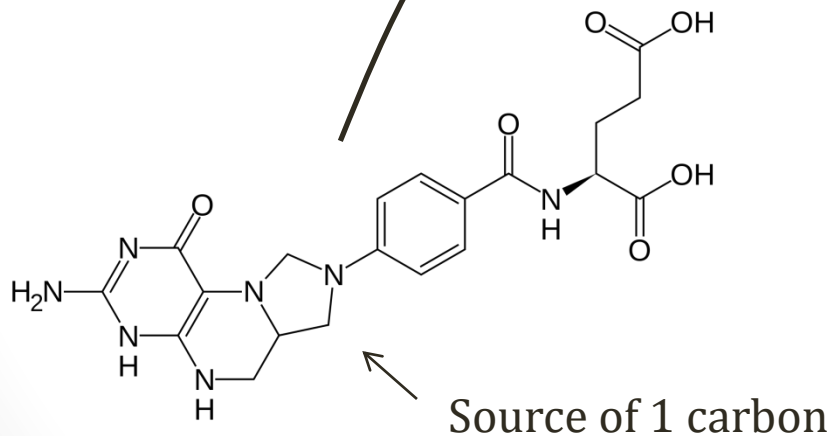
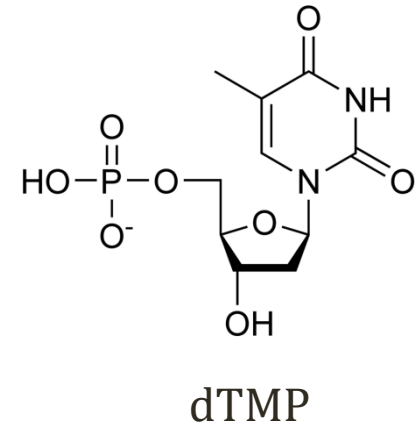


deoxythymidine-MP
(dTMP)

Thymidine



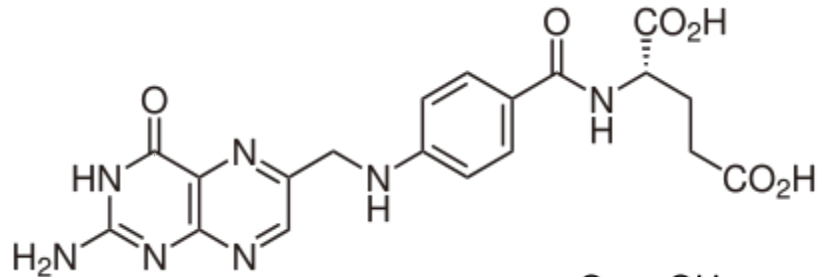
**Thymidylate
Synthase**



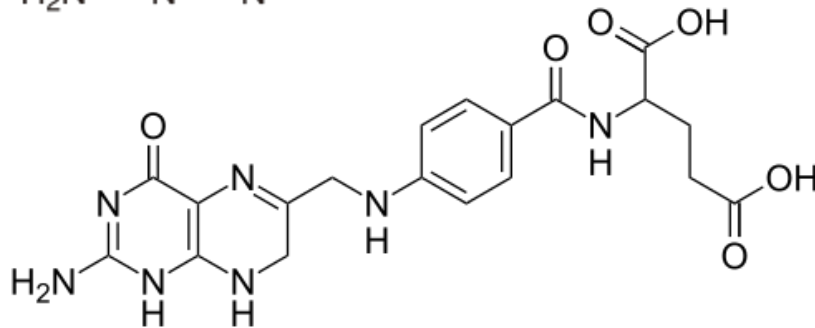
N5, N10 Tetrahydrofolate

Folate Compounds

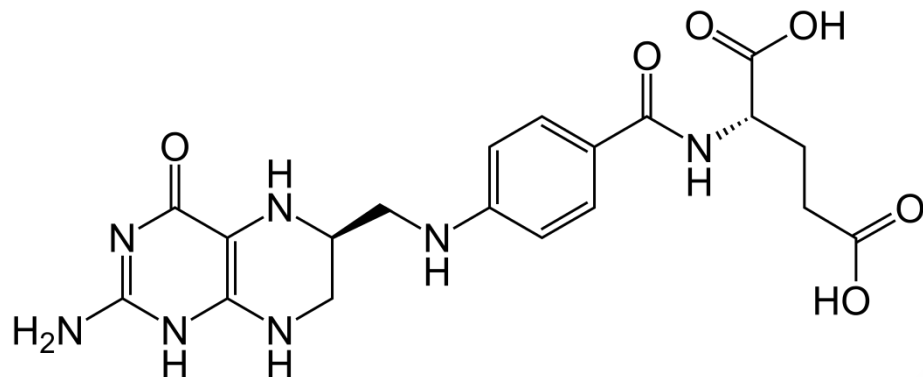
Folate



Dihydrofolate

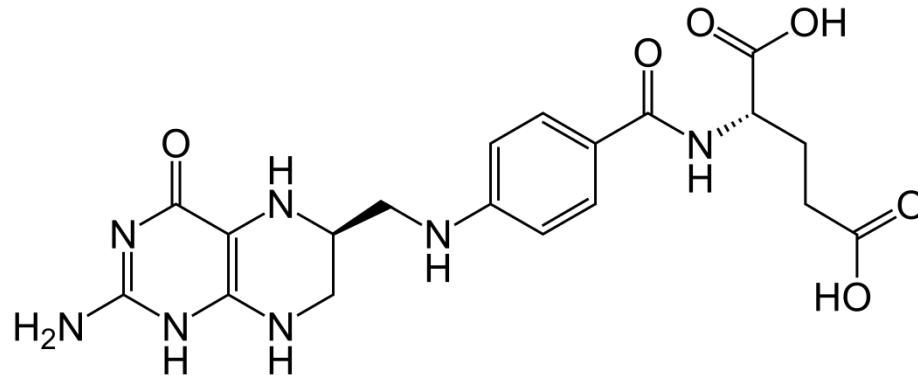


Tetrahydrofolate

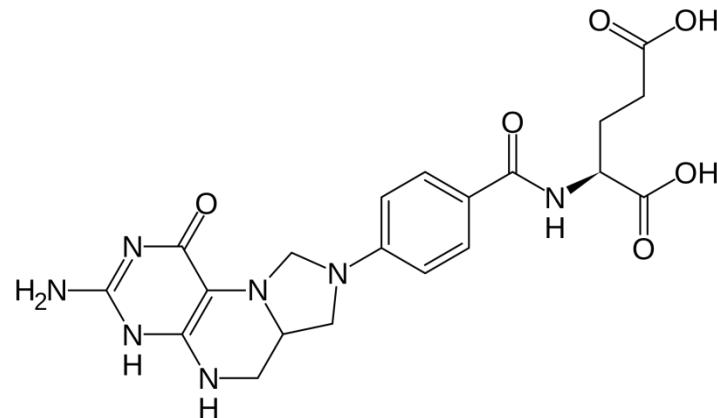


Folate Compounds

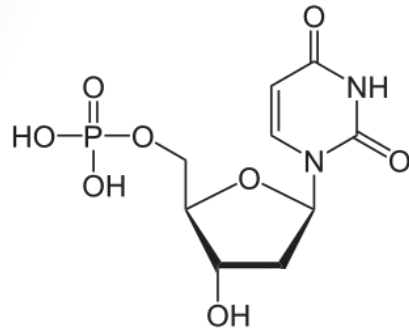
Tetrahydrofolate



N5, N10 Tetrahydrofolate

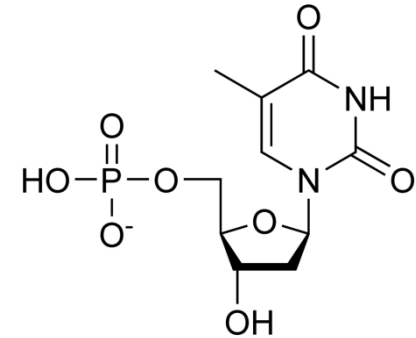


Thymidine



dUridine-MP

**Thymidylate
Synthase**



Thymidine-MP

N5, N10 Tetrahydrofolate

DHF ← Folate

**Dihydrofolate
Reductase**

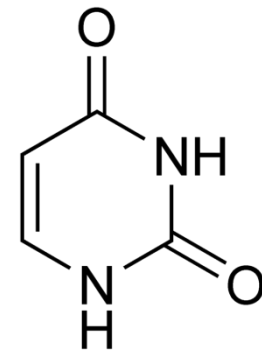
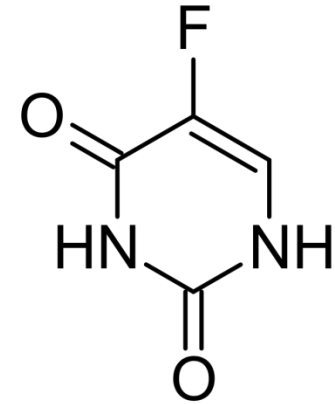
THF

* Folate = 1 carbon carriers

Pyrimidine Synthesis

Drugs and Diseases

- 5-FU
 - Chemotherapy agent
 - Mimics uracil
 - Converted to 5-FdUMP (abnormal dUMP)
 - Covalently binds N5,N10 TFH and thymidylate synthase
 - Result: inhibition thymidylate synthase
 - Blocks dTMP synthesis (“thymineless death”)

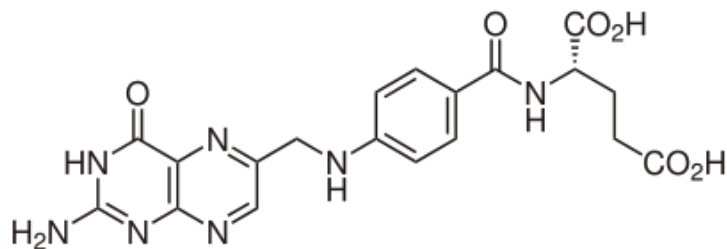


Uracil

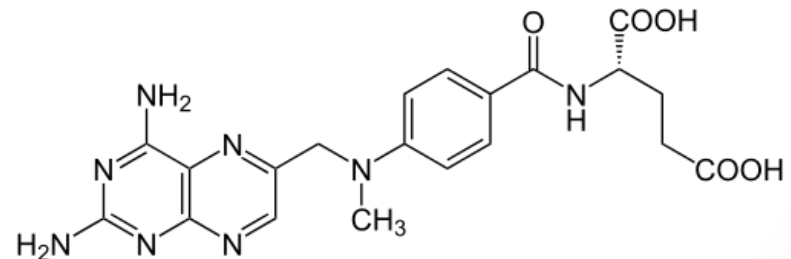
Pyrimidine Synthesis

Drugs and Diseases

- Methotrexate
 - Chemotherapy agent, immunosuppressant
 - Mimics DHF
 - Inhibits dihydrofolate reductase
 - Blocks synthesis dTMP
 - Rescue with leucovorin (folinic acid; converted to THF)



Folate

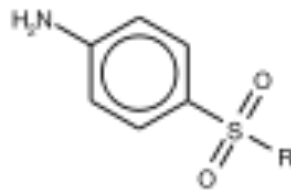


Methotrexate

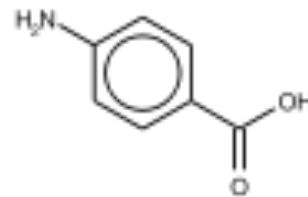
Pyrimidine Synthesis

Drugs and Diseases

- Sulfonamides antibiotics
 - Bacteria cannot absorb folic acid
 - Synthesize THF from para-aminobenzoic acid (PABA)
 - Sulfonamides mimic PABA
 - Block THF synthesis
 - ↓ THF formation → ↓ dTMP (loss of DNA synthesis)
 - No effect human cells (dietary folate)

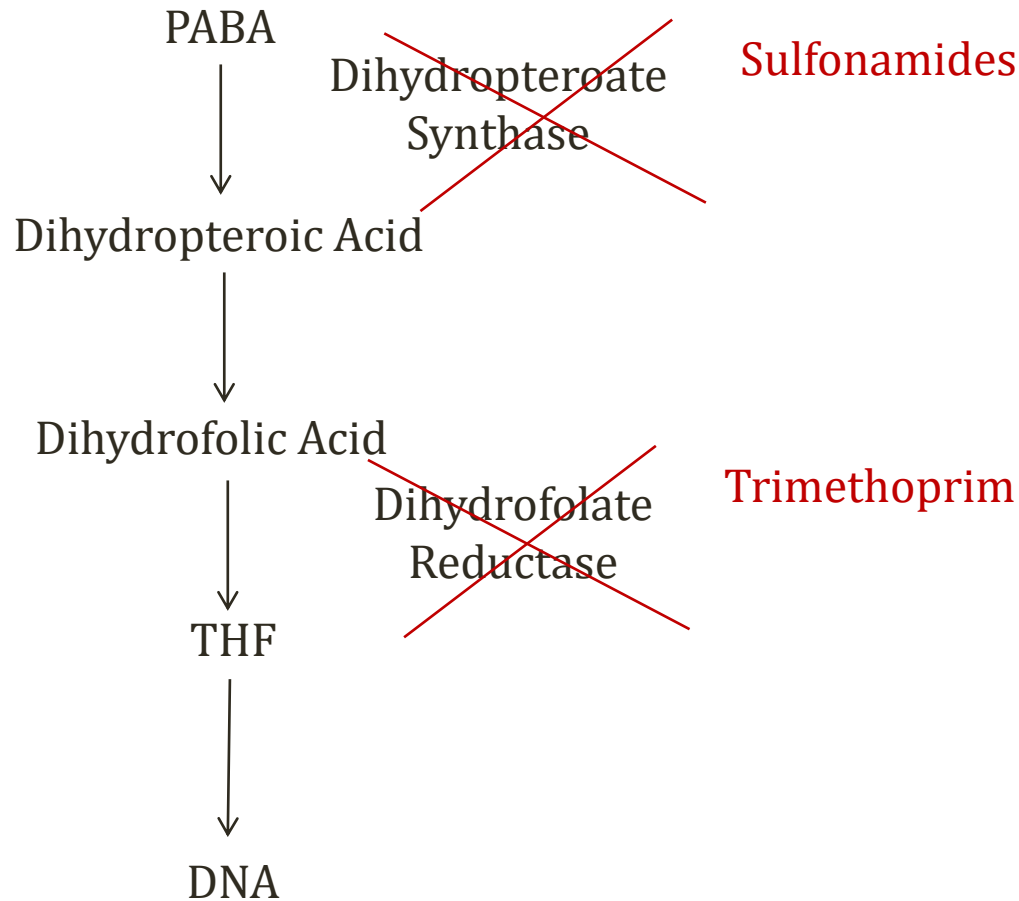


Sulfanilamide



PABA

Bacterial THF Synthesis

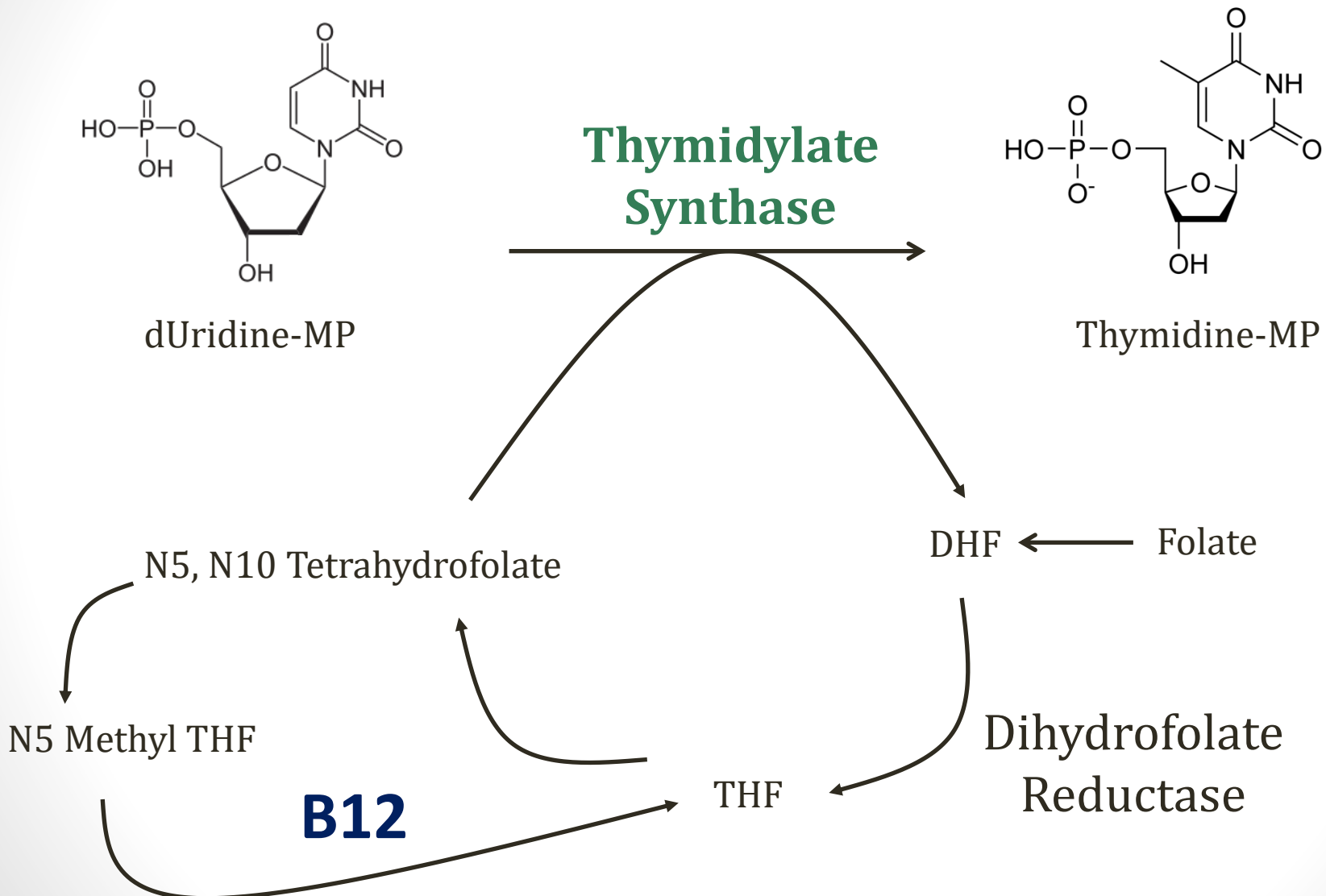


Pyrimidine Synthesis

Drugs and Diseases

- Folate deficiency
 - Main effect: loss of dTMP production → ↓ DNA production
 - RNA production relatively intact (does not require thymidine)
 - **Macrocytic anemia** (fewer but larger RBCs)
 - **Neural tube defects** in pregnancy

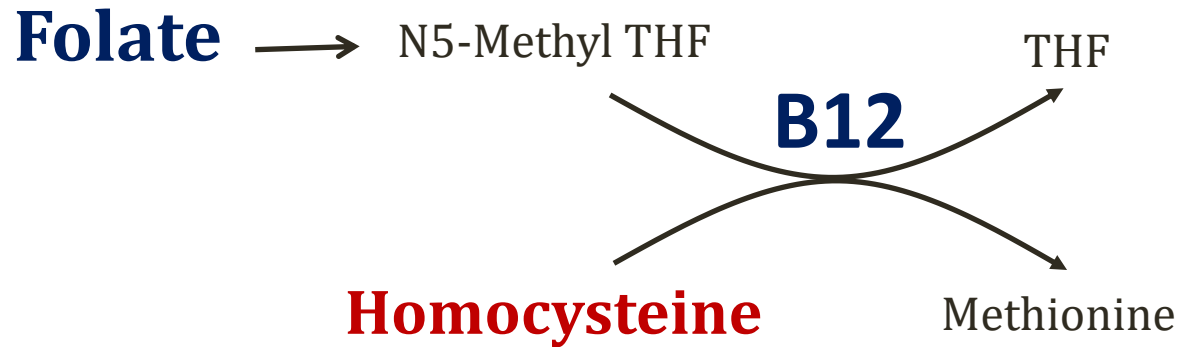
Vitamin B12



Vitamin B12

- Required to regenerate THF from N5-Methyl THF
- Deficiency = “Methyl folate trap”
- Loss of dTMP synthesis (megaloblastic anemia)
- Neurological dysfunction (demyelination)

Homocysteine and MMA



Methylmalonic Acid (MMA)



B12 versus Folate Deficiency

- **Homocysteine**

- Both folate and B12 required to convert to methionine
- Elevated homocysteine in both deficiencies

- **Methylmalonic Acid**

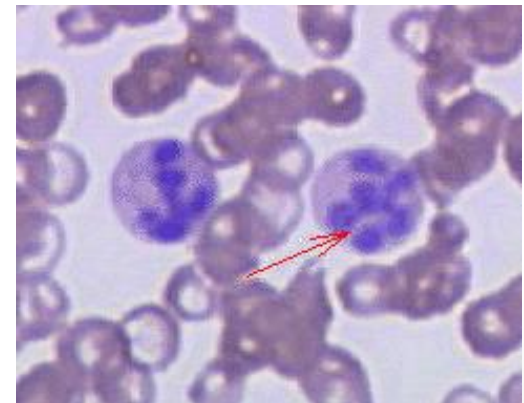
- B12 also converts MMA to succinyl CoA
- B12 deficiency = ↑ methylmalonic acid (MMA) level
- Folate deficiency = normal MMA level

B12 versus Folate Deficiency

	Folate	B12
RBC	↓	↓
MCV	↑	↑
Homocysteine	↑	↑
Methylmalonic acid (MMA)	--	↑

Megaloblastic Anemia

- Anemia ($\downarrow$ Hct)
- Large RBCs ($\uparrow$ MCV)
- Hypersegmented neutrophils
- Commonly caused by defective DNA production
 - Folate deficiency
 - B12 (neuro symptoms, MMA)
 - Orotic aciduria
 - Drugs (MTX, 5-FU, hydroxyurea)
 - Zidovudine (HIV NRTIs)



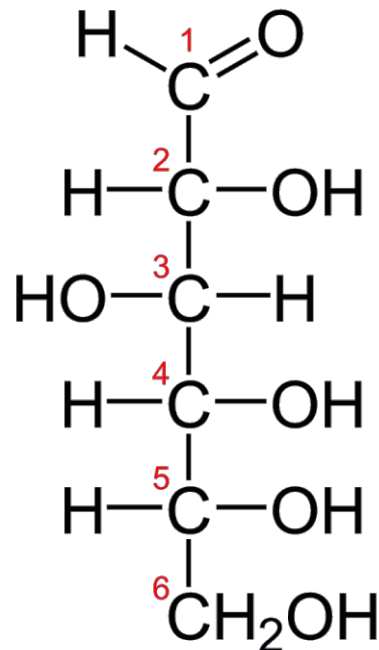
Glucose

Jason Ryan, MD, MPH

Carbs

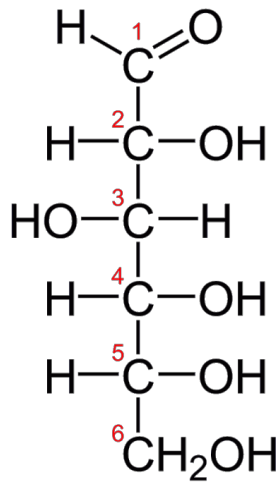
- Carbohydrate = “watered carbon”
- Most have formula $C_n(H_2O)_m$

Glucose
 $C_6H_{12}O_6$

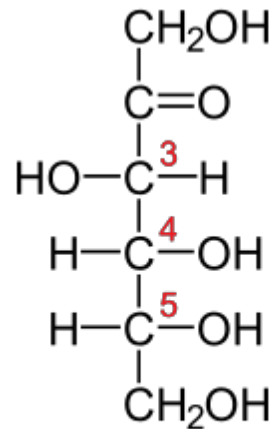


Carbs

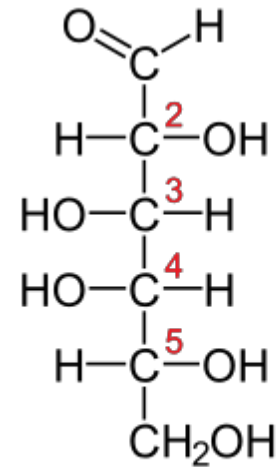
- Monosaccharides ($C_6H_{12}O_6$)
- Glucose, Fructose, Galactose



Glucose



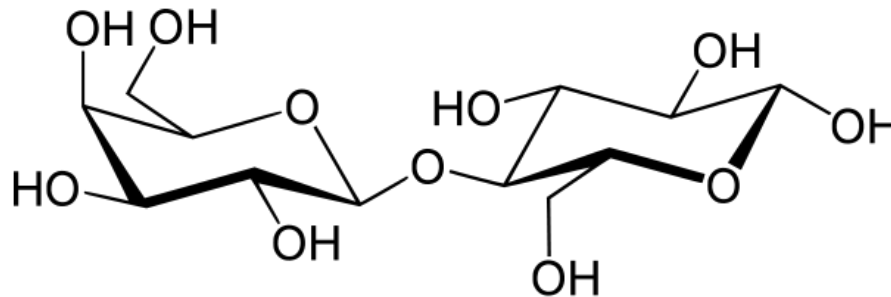
D-Fructose



D-Galactose

Carbs

- Disaccharides = 2 monosaccharides
- Broken down to monosaccharides in GI tract
- **Lactose** (galactose + glucose); lactase
- **Sucrose** (fructose + glucose); sucrase



Lactose

Complex Carbs

- Polysaccharides: polymers of monosaccharides
- Starch
 - Plant polysaccharide (glucose polymers)
- Glycogen
 - Animal polysaccharide (also glucose polymers)
- Cellulose
 - Plant polysaccharide of glucose molecules
 - Different bonds from starch
 - Cannot be broken down by animals
 - “Fiber” in diet → improved bowel function

Glucose

- All carbohydrates broken down into:
 - Glucose
 - Fructose
 - Galactose

Glucose Metabolism

Glucose

**Anaerobic
Metabolism**

TCA Cycle

HMP Shunt

**Fatty Acid
Synthesis**

Glycogenesis

Lactate

H₂O/CO₂

**Ribose/
NADPH**

Fatty Acids

Glycogen

Glucose Metabolism

- Liver
 - Most varied use of glucose
 - TCA cycle for ATP
 - Glycogen synthesis

Glucose Metabolism

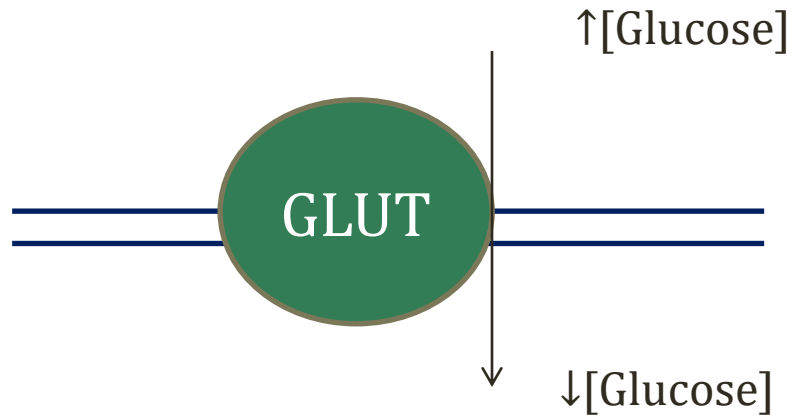
- Brain
 - Constant use of glucose for TCA cycle (ATP)
 - Little glycogen storage
- Muscle/heart
 - TCA cycle (ATP)
 - Transport into cells heavily influenced by insulin
 - More insulin → more glucose uptake
 - Store glucose as glycogen

Glucose Metabolism

- Red blood cells
 - No mitochondria
 - Use glucose for anaerobic metabolism (make ATP)
 - Generate lactate
 - Also use glucose for HMP shunt (NADPH)
- Adipose tissue
 - Mostly converts glucose to fatty acids
 - Like muscle, uptake influenced by insulin

Glucose Entry into Cells

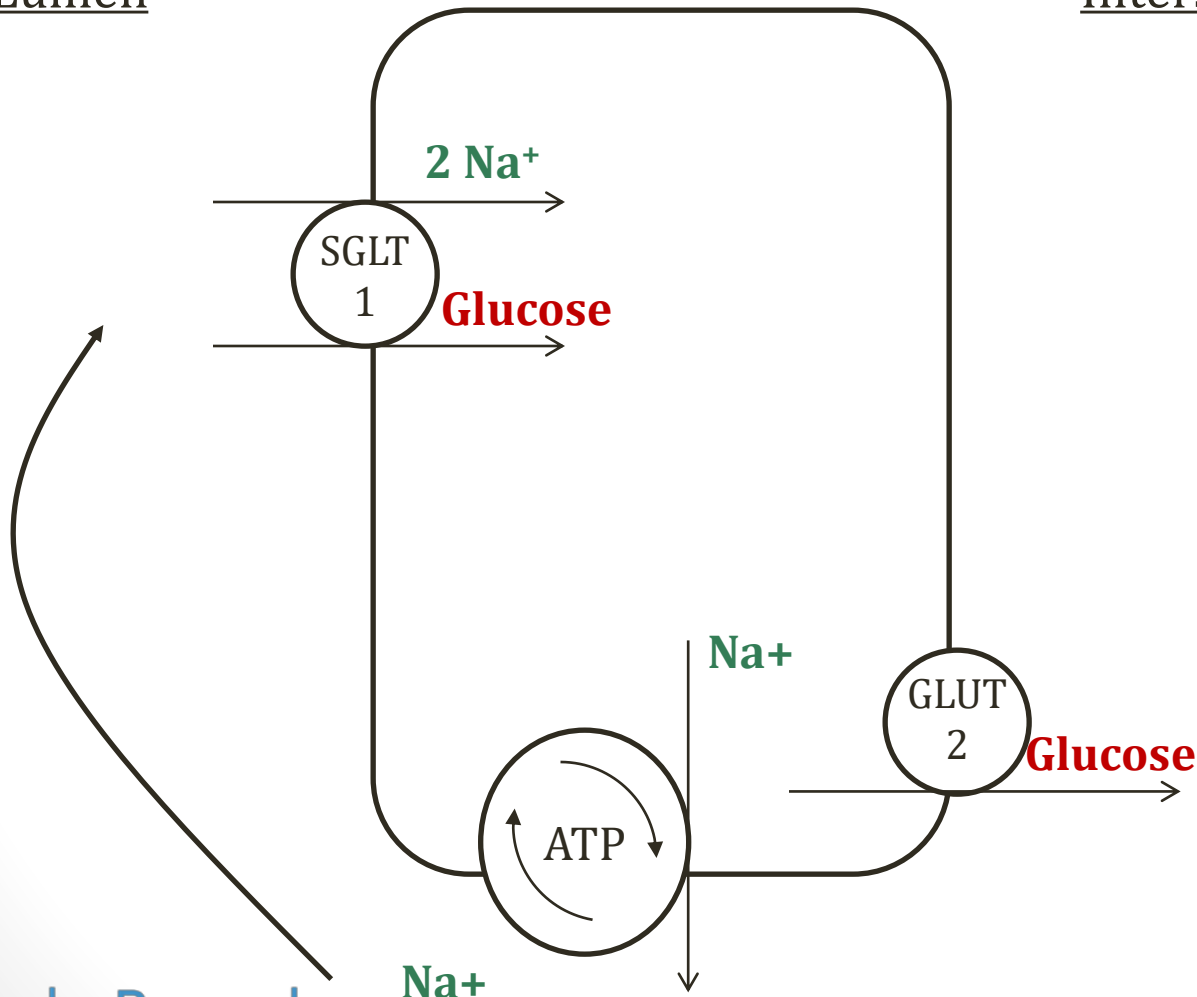
- Na+ **independent** entry
 - 14 different transporters described
 - GLUT-1 to GLUT-14
 - Varies by tissue (i.e. GLUT-1 in RBCs)
- Na+ **dependent** entry
 - Glucose absorbed from low $\rightarrow$ high concentration
 - Intestinal epithelium
 - Renal tubules



Glucose GI Absorption

GI Lumen

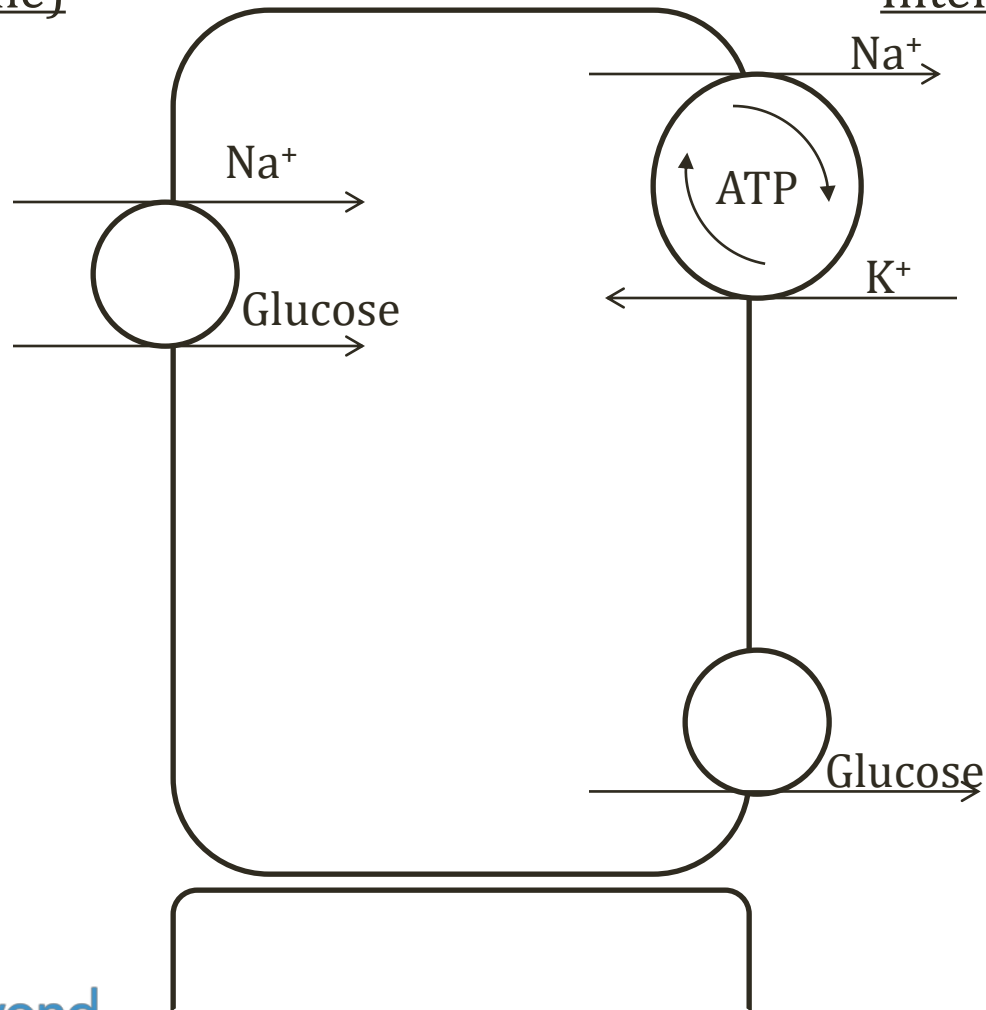
Interstitial/Blood



Proximal Tubule

Lumen (Urine)

Interstitialium/Blood



Glucose Entry into Cells

- GLUT-1
 - Insulin **independent** (uptake when [glucose] high)
 - Brain, RBCs
- GLUT-4
 - Insulin **dependent**
 - Fat tissue, skeletal muscle
- GLUT-2
 - Insulin independent
 - **Bidirectional** (gluconeogenesis)
 - Liver, kidney
 - Intestine (glucose OUT of epithelial cells to portal vein)
 - Pancreas